

**Digital Technology as a Catalyst for Improving Access to Mental Health Services
in Sri Lanka**

(Special Reference to Colombo Sri Lanka)



By

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**Thesis Submitted to the Department of Social Statistics, Faculty of Social Sciences,
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(Hons) in Social Statistics Degree Program.**

Certification of Supervisor

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Declaration of the Student

I declare that the research thesis titled “Digital Technology as a Catalyst for Improving Access to Mental Health Services in Sri Lanka (Special Reference to Colombo Sri Lanka)” was independently compiled under the guidance of Dr. Manjula Gunarathna, Senior lecturer in the Department of Social Statistics, and this thesis was not submitted to complete a degree or course in any other University or Higher Education Institution. I certify that, to the best of my knowledge and belief, there are no prior publications that have not been properly cited.

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Acknowledgment

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Abstract

This study investigates digital technology as a catalyst for improving access to mental health services with special emphasis placed on young adults aged 18-35 in Colombo, Sri Lanka. A structured questionnaire was used in a quantitative research design to collect data from a random sample of young adults in the Colombo metropolitan area. Analysis of the data involved Structural Equation Modeling (SEM) and Chi-Square tests. SEM was employed to examine the relationship between latent variables like Technology Use, Barriers (digital literacy, infrastructure, cost, privacy concerns, stigma), Enabling Factors (mobile penetration, policy support, awareness campaigns, social support, cultural adaptation), and Access to Mental Health Services. The model was very valid and reliable, confirming Technology Use as a mediator, while Enabling Factors and Demographic Characteristics were found to be critical predictors of access to digital mental health services. The Chi-Square tests further demonstrated the presence of significant correlations between personal attributes (age, gender, education, and socioeconomic status) and access to mental health services, which were representative of demographic differences in digital mental health service utilization. The findings are significant for mental health policymakers and stakeholders who seek to promote sustainable and equitable digital mental health solutions for young adults in post-conflict developing nations.

Keywords: *Digital Mental Health, Access to Mental Health Services, Technology Adoption, Barriers and Enabling Factors, Structural Equation Modeling, Chi-Square*

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List of Abbreviations

AMHS	Access to Mental Health Services
AGFI	Adjusted Goodness of Fit Index
AI	Artificial Intelligence
AVE	Average Variance Extracted
BAR	Barriers
CBT	Cognitive Behavioral Therapy
CFA	Confirmatory Factor Analysis
CFI	Comparative Fit Index
CMIN	Chi-Minimum (Chi-square statistic)
COVID	Coronavirus Disease
CR	Composite Reliability / Critical Ratio
dCBT	Digital Cognitive Behavioral Therapy
DF	Degrees of Freedom
DMH	Directorate of Mental Health
DTA	Digital Technology Adoption
eHealth	Electronic Health
eMHMIS	Electronic Mental Health Management Information System
ENF	Enabling Factors
GFI	Goodness of Fit Index
IT	Information Technology
LKR	Sri Lankan Rupee
LMIC(s)	Low- and Middle-Income Countries
MHPSS	Mental Health and Psychosocial Support
mHealth	Mobile Health
MLE	Maximum Likelihood Estimation
NFI	Normed Fit Index
NIMH	National Institute of Mental Health
PCFI	Parsimony Comparative Fit Index
PGFI	Parsimony Goodness of Fit Index

PNFI	Parsimony Normed Fit Index
PRATIO	Parsimony Ratio
RFI	Relative Fit Index
RMR	Root Mean Square Residual
RMSEA	Root Mean Square Error of Approximation
SEM	Structural Equation Modeling
SMS	Short Message Service
TLI	Tucker-Lewis Index
TRCSL	Telecommunications Regulatory Commission of Sri Lanka
WHO	World Health Organization
TRCSL	Telecommunications Regulatory Commission of Sri Lanka

Chapter One: Introduction

1.1 Background of study

Sri Lanka faces a significant mental health burden resulting from decades of conflict, natural disasters, and economic crises. The nation experienced four decades of violent conflict, a devastating tsunami, terrorist attacks including the 2019 Easter Sunday bombings, and a severe economic crisis beginning in 2022 (Deivanayagam et al., 2024). People living with mental health difficulties experience individual, interpersonal, social and structural barriers to help-seeking, including stigma, lack of knowledge, denial, fear of societal repercussions, language barriers, and financial constraints (Deivanayagam et al., 2024). During the past several decades, Sri Lanka has seen rapid development of psychiatry services throughout the country, with services becoming decentralized and extended to all districts, though much remains to be done in light of recent economic and socio-political upheavals (Amarasinghe & Rajapakse, 2024).

The digital health infrastructure in Sri Lanka has expanded considerably, creating new opportunities for mental health service delivery. Compared with other less economically developed countries, Sri Lanka has maintained a high standard of health care nationwide, though urban hospitals have more up to date facilities and professional care than rural areas (Jayasinghe et al., 2016). Telephone usage in Sri Lanka has exceeded 95%, presenting opportunities to provide more healthcare despite insufficient qualified staff. Telemedicine and telehealth are integral parts of healthcare in many developed countries, though Sri Lanka is yet to fully harness their potential, with various solutions having emerged in an ad hoc manner over the past decade (Marasinghe et al., 2020). This widespread connectivity has created a foundation for implementing digital mental health interventions that can reach populations previously underserved by traditional healthcare infrastructure.

The COVID-19 pandemic served as a critical catalyst for digital mental health adoption in Sri Lanka, accelerating initiatives that had been developing slowly prior

to the crisis. The Directorate of Mental Health (DMH) within the Ministry of Health prioritized delivery of Mental Health and Psychosocial Support (MHPSS), taking proactive preventive measures combined with an effective public health approach (World Health Organization, 2020). WHO provided seamless support to DMH through strategic assistance, capacity building, coordination and technical assistance (World Health Organization, 2020). Wellbeing messages were inserted as ring tones in mobile phone lines during the peak of the pandemic as an innovative mass communication approach. The unprecedented rise in COVID-19 cases since July 2021 led to establishment of an integrated patient management system where patients could access medical professionals through SMS and calling, providing telephone triage, patient advice, and coordination with the national ambulance system (Karunathilake et al., 2022). These rapid implementations demonstrated both the potential and challenges of scaling digital mental health services in resource limited settings.

Telemedicine and telepsychiatry have emerged as important modalities for expanding access to mental health care in Sri Lanka, though implementation has encountered significant obstacles. Telepsychiatry, defined as the use of electronic communication and information technologies to provide or support clinical psychiatric care at a distance, stands to benefit mental health services as physical examination may not always be necessary to arrive at a diagnosis (Perera et al., 2020). While telepsychiatry has been in practice in several countries for many years and there is growing evidence showing its success in a range of psychological problems, it is relatively new to South Asia and Sri Lanka (Perera et al., 2020). In a study assessing the impact of the pandemic on treatment compliance at a child and adolescent mental health service, sixty percent of patients defaulted treatment during lockdowns, with rates significantly higher for those receiving only nonpharmacological therapies (Rohanachandra & Nanayakkara, 2020). Key challenges include lack of sophisticated infrastructure and internet connectivity impeding access for rural areas, with most adults being computer illiterate digital immigrants facing obstacles related to cost, accessibility, and ease of use (Perera et al., 2020). These barriers underscore the need

for comprehensive capacity building and infrastructure development to support sustainable telepsychiatry services.

WHO is working with DMH and Sri Lanka Telecom to build capacity and decentralize this helpline service to district level to accommodate escalating distressed calls (World Health Organization, 2020). Research has demonstrated that mHealth interventions using mobile apps and SMS messaging show promising efficacy, with studies revealing significant reductions in anxiety, stress, and depression (Rathbone & Prescott, 2017). Just-in-time adaptive mHealth interventions that execute decision rules at pre-specified decision points can provide precision support adapted to individuals' changing circumstances, initiated through push mechanisms like SMS and notifications without requiring individuals to remember to access interventions (Bidargaddi et al., 2020). These targeted interventions demonstrate how mobile technology can be leveraged to address specific mental health needs across different populations and settings.

Digital stress management interventions and mental health applications represent another frontier in Sri Lanka's digital mental health landscape, with growing attention to workplace mental health and accessible therapy options. Research on digital stress management interventions has revealed user preferences for platforms that integrate personal and collaborative features with engagement enhancing elements, suggesting that user experience is a critical factor in adoption. Online therapy platforms have emerged offering personalized therapy tailored to Sri Lankan cultural contexts, with licensed therapists providing cognitive behavioral therapy and other evidence-based approaches through secure video sessions, chat, or phone (TherapyMantra, 2023).

Despite promising developments, significant barriers and challenges continue to impede widespread adoption of digital health technologies for mental health in Sri Lanka. Lack of necessary IT infrastructure, knowledge of applications, and the usefulness of e-technology, along with absence of national policy and legal frameworks to introduce e-health solutions, represent major obstacles (Jayasinghe et al., 2016). A model exploring factors characterizing introduction of telemedicine to

rural areas identified culture, technology, and infrastructure as critical factors that hinder acceptance (Jayasinghe et al., 2016). The country did not have regulations or guidelines specific to telemedicine services when the pandemic started, prompting the Health Information Unit of the Ministry of Health to develop non-formal guidelines for voluntary participating doctors (Marasinghe et al., 2020). Issues during telehealth planning and implementation include securing adequate funds, limited knowledge of information technology among health staff, reluctance of patients to use video consultation, difficulties in allocating responsibility, and network facility provision to peripheral hospitals (Kumarasinghe et al., 2022). These multifaceted challenges require coordinated responses involving technology development, digital literacy programs, policy reform, and community engagement.

In conclusion, the evidence demonstrates that digital technology holds significant promise for improving access to mental health services in Sri Lanka, particularly among young adults in urban centers such as Colombo. The country's expanding digital infrastructure, combined with high smartphone penetration among young adults and growing policy support, provides a foundation for leveraging digital technology to address critical gaps in mental health service access. However, persistent barriers including digital literacy constraints, infrastructure limitations, privacy concerns, and stigma continue to impede utilization of digital mental health services, while enabling factors such as mobile penetration and social support networks present opportunities to enhance access. Furthermore, demographic factors including age, gender, educational attainment, and socioeconomic status are likely to influence the level of access to mental health services, underscoring the need for targeted and equity-sensitive approaches. Understanding the structural relationships between technology use, barriers, enabling factors, and access outcomes among young adults in Colombo is therefore essential for developing evidence-based digital mental health strategies that are contextually appropriate, equitable, and sustainable.

1.2 Statement of the problem

Mental health issues in Sri Lanka have grown significantly over the past few decades, and have become a major national problem with severe social, economic, and health impacts. According to the World Health Organization, a large portion of the world's population suffers from various mental health conditions, but in developing countries, many of these patients are unable to access the necessary treatment due to lack of resources, lack of trained mental health professionals, and stigmatization of mental illness in society (World Health Organization, 2022). This problem is also acute in Sri Lanka, where access to mental health services is limited, especially for people living in rural and remote areas, and many patients are unable to receive timely treatment (Ministry of Health Sri Lanka, 2021).

The current mental health care system in Sri Lanka is largely hospital-based, and patients face many challenges, including long wait times to see specialists, high travel costs, and disruption to employment and education. In addition, the stigma associated with mental illness has led many people to hide their problems or hesitate to seek treatment (Perera et al., 2020). This situation has further exacerbated the disease and has increased social and economic consequences.

In this context, developing mental health services in a more efficient, accessible, and modern manner through digital technology can be identified as an important solution. Digital systems, especially telepsychiatry, e-Health and electronic mental health management information systems (eMHMIS), have been shown to be more efficient in providing remote medical consultations, systematically managing patient medical information, monitoring treatment progress and planning health services (DIAL, 2023). With the rise of telemedicine services, especially during the COVID-19 pandemic, the importance of digital mental health services has been further confirmed (WHO, 2021).

However, there are still various challenges for the full implementation of digital mental health services in Sri Lanka. Among them, the lack of internet and technological infrastructure, limited digital knowledge of health staff, legal issues

regarding data confidentiality and privacy, and lack of trust in technological systems are the main ones (Perera et al., 2020). Therefore, there is an urgent need to examine how digital technology can effectively improve access to mental health services among young adults in Sri Lanka, particularly in Colombo. This requires identifying the key factors and structural relationships between technology use, barriers, and enabling factors that determine access; assessing the prevalence of those barriers and enabling factors affecting digital mental health service utilization; and understanding how demographic characteristics such as age, gender, education, and income influence the level of access to mental health services. Addressing these knowledge gaps will provide evidence-based insights to support the development of effective, equitable, and contextually appropriate digital mental health strategies for Sri Lanka.

1.3 Research Questions

1.3.1 Key Question

What factors influence access to mental health services through digital technology among young adults?

1.3.2 Sub Questions

- What is the relationship between demographic factors and the level of access to mental health services among young adults in Sri Lanka?
- What is the prevalence of barriers and enabling factors affecting the use of digital mental health services among young adults in Sri Lanka?

1.4 Research Objectives

1.4.1 Key Objective

Identifying and evaluating factors influencing access to mental health services through digital technology among young adults.

1.4.2 Specific objectives

- To examine the relationship between demographic factors and the level of access to mental health services among young adults in Sri Lanka.
- To assess the prevalence of barriers and enabling factors affecting the use of digital mental health services among young adults in Sri Lanka.

1.5 Significance of the Study

Mental health has become a growing public health concern in Sri Lanka, with increasing rates of depression, anxiety, stress-related disorders, and suicide, particularly among young adults. Despite the high burden of mental health conditions, access to mental health services remains severely limited, with approximately 85–90% of Sri Lankans with mental health conditions receiving no professional treatment (Ministry of Health Sri Lanka, 2023). This study is significant because it directly addresses this critical access gap by investigating how digital technology can serve as a catalyst for improving access to mental health services among young adults in Colombo, where digital infrastructure is most developed and the potential for scalable digital solutions is greatest.

Sri Lanka faces a critical shortage of psychiatrists, psychologists, and mental health counselors, particularly outside major cities (Ministry of Health Sri Lanka, 2021). Many individuals are unable to receive timely care due to long waiting times, distance, and lack of specialized professionals. This study is significant because it explores how digital technologies such as telepsychiatry, mobile health (mHealth) applications, and online counseling platforms can help overcome workforce shortages and expand mental health service coverage.

Digital health technologies have transformed healthcare delivery worldwide by improving access, efficiency, and patient engagement (WHO, 2021). Mobile applications, telepsychiatry, and online counseling platforms have demonstrated potential in extending mental health service reach beyond physical and temporal constraints. This study is significant as it goes beyond describing these technologies to empirically examining the structural relationships between technology use, barriers, and enabling factors, thereby identifying the key factors that determine whether digital technology translates into improved access for young adults in Sri Lanka's specific socio-cultural context.

Despite growing availability of digital mental health technologies, significant barriers continue to prevent young adults in Sri Lanka from accessing these services, while

certain enabling factors have the potential to facilitate uptake. This study is significant because it systematically assesses the prevalence of both barriers (such as digital literacy gaps, cost, privacy concerns, and stigma) and enabling factors (such as mobile penetration, social support, and government policy) that affect the use of digital mental health services among young adults in Colombo. Generating evidence on the relative prevalence of these factors is essential for designing targeted interventions that amplify enablers and address the most critical barriers to access.

Mental health stigma remains a major obstacle preventing individuals from seeking professional help in Sri Lanka (Fernando & Weerackody, 2019). Digital technologies offer anonymity and privacy, encouraging individuals to seek help without fear of social judgment. This study is significant because it demonstrates how online platforms and mobile applications can reduce stigma and promote early intervention.

Young adults aged 18-35 represent a digitally active population with high rates of smartphone ownership and internet use, making them an important target population for digital mental health interventions (Telecommunications Regulatory Commission of Sri Lanka, 2023). At the same time, this age group experiences high rates of depression, anxiety, and stress-related conditions, yet faces unique barriers to accessing professional mental health care including stigma, limited awareness, and financial constraints. This study is significant because it focuses specifically on this population, investigating the factors and demographic characteristics that either facilitate or impede their access to mental health services through digital technology in the Colombo context.

Traditional mental health services often require high operational costs, infrastructure, and human resources. Digital interventions offer a cost-effective alternative by reducing travel expenses, consultation costs, and hospital overcrowding (Naslund et al., 2017). This study highlights the economic benefits of digital mental health solutions, making them particularly valuable for a developing country like Sri Lanka with limited healthcare budgets.

Demographic factors such as gender, age, educational level, income, and employment status are known to influence health-seeking behavior and technology adoption patterns. However, there is limited evidence on how these demographic characteristics specifically shape the level of access to mental health services through digital technology among young adults in Sri Lanka. This study is significant as it examines these demographic relationships, providing policymakers with evidence to design demographically targeted digital mental health strategies and to ensure that digital mental health services are equitable and inclusive across diverse segments of Colombo's young adult population.

There is limited academic literature focusing specifically on the factors determining access to digital mental health services among young adults in the Sri Lankan context. Existing studies rarely examine structural relationships between technology use, barriers, enabling factors, and access outcomes, and few have analyzed the role of demographic factors in shaping access levels. This study fills these important research gaps by providing localized empirical evidence on structural determinants of access, the prevalence of barriers and enabling factors, and the influence of demographic characteristics. The findings will contribute to regional and global research on digital mental health access in low- and middle-income countries, and will directly inform evidence-based policy and program development to improve mental health service access for young adults in Sri Lanka, supporting long-term social and economic development goals.

1.6 Limitation of the study

Despite the contributions of this study, several limitations should be acknowledged.

The study is restricted to the Colombo metropolitan area, which is highly urbanized and has better digital infrastructure compared to other parts of Sri Lanka. While this setting allows for easy access to technology and a concentrated population of young adults, it limits the applicability of findings to rural or semi-urban populations. Youth in other regions may face different socio-economic conditions, cultural contexts, and

levels of access to digital mental health tools, making it difficult to generalize results nationally.

This research specifically targets young adults between 18 and 35 years. While this group represents a digitally active and socially mobile segment of the population, the study's findings may not reflect the experiences of adolescents under 18 or older adults, who may engage with technology differently and experience distinct mental health challenges. Consequently, the

By focusing solely on Colombo, the study excludes estate-sector populations, who often face different socio-economic and educational challenges. These populations may have limited internet access, lower smartphone ownership, and fewer digital literacy skills. Excluding them narrows the study's scope and prevents insights into disparities in digital mental health engagement between urban and estate-sector youth.

The study relies primarily on self-reported information collected through surveys, interviews, or questionnaires. Participants may not always accurately report their mental health status or digital usage due to forgetfulness, social desirability, or fear of stigma. This reliance on self-reporting can introduce bias and affect the reliability of the data.

While this study examines the structural relationships between technology use, barriers, enabling factors, and access to mental health services, the cross-sectional nature of the research design limits the ability to establish causal directionality among these variables. The structural relationships identified reflect associations at a single point in time and may not fully capture dynamic or evolving patterns of digital mental health access. Longitudinal studies would be required to confirm causal pathways and track how access levels change as barriers are addressed or enabling factors strengthen.

Even within Colombo, not all young adults have equal access to digital devices, high-speed internet, or modern smartphones. Differences in socio-economic status,

employment, or education may affect participation and engagement with digital interventions. This limitation could create variability in the study's outcomes, as some participants may be underrepresented or unable to fully utilize the tools provided.

The study is conducted over a limited time period, which may not capture long term impacts of digital interventions on mental health. Behavioral and emotional improvements often require sustained use of tools or repeated interventions. A short-term evaluation may provide insights into immediate engagement and satisfaction but may fail to measure sustained psychological benefits or long-term behavior change.

The study targets non clinical young adults, excluding those with diagnosed mental health disorders. While this allows the study to focus on access patterns among the general young adult population, which is the primary group targeted by digital mental health access initiatives, it limits the ability to assess whether barriers, enabling factors, and demographic relationships with access differ for individuals with diagnosed conditions, who may have distinct service utilization needs and patterns.

Participants' familiarity and comfort with digital technology vary widely. Some may be adept at navigating apps and online resources, while others may struggle with basic functionalities. These differences in digital literacy could influence both the engagement with digital interventions and the effectiveness of these tools in improving mental health outcomes.

Mental health remains a sensitive and stigmatized topic in Sri Lanka. Participants may provide responses that they perceive as socially acceptable rather than fully honest, particularly regarding their mental health experiences or technology usage. This can lead to underreporting of psychological distress or overreporting of positive behaviors, affecting the study's validity.

Digital platforms, apps, and online interventions evolve rapidly. A tool that is popular or effective at the time of the study may become obsolete, unsupported, or replaced by new technologies shortly afterward. This rapid evolution poses a limitation for the long-term relevance and applicability of the study's findings.

The study may rely primarily on self-reported wellbeing scores, app usage statistics, or survey responses to measure the level of access to mental health services. Self-reported access measures may not fully capture objective dimensions of access such as actual service utilization rates, wait times, or quality of care received. Future studies incorporating objective access metrics or administrative health data would strengthen the validity and comprehensiveness of findings on digital technology's role in improving mental health service access.

Despite Colombo's urban environment, cultural stigma around mental health persists. Young adults may avoid openly discussing psychological difficulties or may hesitate to engage fully with digital mental health interventions. These barriers can affect both participation rates and the authenticity of responses.

Digital tools and interventions may be primarily available in English or Sinhala, excluding Tamil-speaking youth or bilingual participants who may find it harder to use or understand the content. Language limitations can reduce engagement and accessibility, particularly in a multi-lingual urban context like Colombo.

The adoption and implementation of digital mental health tools depend on institutional support, health policies, and regulatory approvals. Limitations in government or organizational backing, digital infrastructure, or mental health policy alignment can restrict the scalability and sustainability of digital interventions, limiting the long-term impact of the study.

1.7 Chapter Outline

The first chapter introduces the research topic and provides the background on the role of digital technology in improving access to mental health services among young adults aged 18–35 in Colombo, Sri Lanka. It outlines the key research problem and establishes three core objectives: identifying structural relationships between technology use, barriers, and enabling factors affecting access; assessing the prevalence of barriers and enabling factors; and examining the relationship between demographic factors and the level of access to mental health services. This chapter

also presents the research questions, significance, scope, and limitations of the study, setting a clear foundation for the research.

Chapter two reviews existing literature on digital technology and access to mental health services, with particular emphasis on barriers and enabling factors, the Technology Acceptance Model, and the role of demographic factors in service access. It examines global and local studies, identifies research gaps relevant to young adults in Sri Lanka, and discusses the theoretical frameworks underpinning the study's three objectives. This chapter establishes the conceptual context and supports the formulation of the research methodology.

The third chapter explains the research design, sampling methods, data collection techniques, and analytical tools used in the study. It details the selection of participants, including their age range and location within Colombo, and discusses ethical considerations and data validity to ensure transparency and reliability in the research process.

Chapter four presents and analyzes the findings of the study in line with the three research objectives. It reports on the structural relationships between technology use, barriers, and enabling factors; the prevalence of barriers and enabling factors among the sample; and the relationships between demographic characteristics and levels of access to mental health services. Data are systematically examined using appropriate statistical methods, with results presented through tables, charts, and structural models where relevant.

The final chapter interprets the results in relation to the research objectives and literature. It discusses the implications of the findings for digital mental health interventions, identifies study limitations, and provides practical recommendations for policymakers, healthcare providers, and technology developers. The chapter concludes with suggestions for future research to enhance mental health support for Colombo's youth.

Chapter Two: Literature Review

2.1 Introduction

The purpose of this literature review is to provide a comprehensive overview of existing research on digital technology's role in improving access to mental health services, with particular emphasis on low- and middle-income country (LMICs) contexts and Sri Lanka specifically. This chapter synthesizes key themes, theories, and empirical findings that emerge from contemporary research on digital mental health, technology adoption, barriers to access, and youth engagement with mental health services. Through this extensive literature review, the following key points are highlighted: the concept and scope of digital mental health, global trends in digital mental health implementation, the mental health landscape in Sri Lanka, digital technologies used in mental health service delivery, barriers to digital mental health services, enabling factors for digital adoption, access to mental health services among young adults, and the intersection of digital technology and mental health outcomes. This chapter integrates global, regional, and local studies to justify the current investigation and identify research gaps.

2.2 The Concept of Digital Health and Mental Health Services

This section defines digital health (eHealth/mHealth) as the use of digital tools to support healthcare delivery. It then narrows into digital mental health, covering apps, telepsychiatry, online counseling, and digital cognitive behavioral therapy (dCBT) as ways to improve mental health outcomes.

2.2.1 Digital Health

Digital health, also referred to as eHealth or mHealth (mobile health), encompasses the use of digital technologies, information systems, and communication tools to support health service delivery and health-related activities (World Health Organization, 2019). According to the World Health Organization, digital health includes electronic health records, telemedicine, mobile health applications, health information systems, and other technology-enabled health interventions. The scope of digital health extends beyond simple digitization of paper records; it represents a

fundamental shift in how health services are conceptualized, delivered, and accessed by populations (Eysenbach, 2001). Digital health technologies leverage smartphones, computers, the internet, and mobile applications to bridge geographical, temporal, and resource constraints that traditionally limit healthcare access (Aranda Jan et al., 2014).

In the context of developing economies, digital health has emerged as a critical strategy for addressing healthcare capacity constraints, extending service reach to underserved populations, and improving the efficiency of limited health resources (Labrique et al., 2018). Digital health interventions are particularly valuable in resource-limited settings because they can reduce the burden on overstretched health systems while simultaneously improving the timeliness and accessibility of care (Blaya et al., 2010).

2.2.2 Digital Mental Health

Digital mental health, a subset of digital health, refers to the application of digital technologies and platforms to improve mental health outcomes, increase access to mental health services, and support mental health promotion, prevention, and treatment (Naslund et al., 2017). Digital mental health encompasses a diverse range of interventions including mobile applications for symptom monitoring and self-management, telepsychiatry (remote psychiatric consultations via video or telephone), online counseling platforms, dCBT, peer support networks, and mental health information websites (Riva et al., 2016; Torous et al., 2019).

The global mental health crisis, characterized by the substantial burden of depression, anxiety, and other mental health conditions, has created an urgent need for innovative service delivery models. The World Health Organization reports that approximately 280 million people worldwide suffer from depression, yet access to treatment remains severely limited, particularly in low- and middle-income countries where the treatment gap exceeds 80% (WHO, 2022). Digital mental health represents a scalable, cost-effective approach to narrowing this treatment gap by enabling individuals to access evidence-based interventions without requiring in-person contact with specialized mental health professionals (Naslund et al., 2017; Alagarajah et al., 2024).

2.3 Global Trends in Digital Mental Health Implementation

Covers how high-income countries rapidly expanded digital mental health, especially after COVID-19. Also discusses how LMICs have adopted simpler tools like SMS-based interventions with promising results

2.3.1 Adoption of Digital Mental Health Technologies Worldwide

The global adoption of digital mental health has accelerated rapidly, particularly since the COVID-19 pandemic, which necessitated rapid expansion of remote mental health services (Gale et al., 2022). Telemedicine and digital mental health platforms have expanded dramatically across developed nations, with countries such as the United States, Australia, and several European nations implementing comprehensive digital mental health strategies and funding mechanisms to support technology-enabled care (Torous et al., 2019; Naslund et al., 2017).

In high-income countries, digital mental health adoption has been driven by a combination of factors: integration of telehealth into insurance reimbursement systems, increasing smartphone penetration, technological infrastructure development, and growing demand from digitally literate populations seeking convenient, stigma-reduced access to mental health services (Gale et al., 2022). Studies conducted in the United States and United Kingdom have documented substantial increases in telemental health utilization, with mental health professionals reporting increased comfort and competence in delivering care through digital platforms (Pierce, 2020). Mobile mental health applications in developed nations have proliferated, with hundreds of applications now available through mainstream app stores, offering functions ranging from mood tracking and guided meditation to cognitive behavioral therapy modules (Torous et al., 2017).

2.3.2 Digital Mental Health in Low- and Middle-Income Countries (LMICs)

The expansion of digital mental health in LMICs has been more gradual and uneven, constrained by infrastructure limitations, limited digital literacy, inconsistent internet connectivity, and funding constraints (Naslund et al., 2017; Alagarajah et al., 2024). However, emerging evidence from South Asia, Sub Saharan Africa, and Southeast Asia demonstrates that even in resource-limited settings, digital mental health approaches

can be effectively implemented and adapted to local contexts (Aranda-Jan et al., 2014; Senanayake & Senanayake, 2017).

Research from South Asian countries including India, Pakistan, and Bangladesh has shown that mobile based mental health interventions, often utilizing simple SMS (short message service) technology rather than sophisticated applications, can effectively reach large populations and improve mental health outcomes (Aranda Jan et al., 2014). A study conducted across multiple South Asian countries documented that text message-based interventions for depression and anxiety were feasible, acceptable, and associated with significant reductions in symptom severity among recipients (Naslund et al., 2017). These findings suggest that digital mental health approaches need not rely on advanced technology infrastructure; even basic digital tools can yield meaningful improvements in mental health outcomes in LMIC contexts.

2.4 Mental Health Services and Access: Global and Sri Lankan Contexts

Explains the global treatment gap (75-85% of people go untreated). For Sri Lanka specifically, it highlights that 85-90% of those with mental health conditions receive no treatment, with services heavily concentrated in Colombo.

2.4.1 The Global Mental Health Crisis and Treatment Gap

Mental health disorders represent a major contributor to the global burden of disease, with depression, anxiety disorders, and other mental conditions affecting hundreds of millions of people worldwide (WHO, 2022). Despite the substantial prevalence of mental health conditions, access to treatment remains profoundly limited. The global treatment gap for mental health conditions is estimated at 75-85%, meaning that the vast majority of individuals suffering from mental health conditions do not receive any professional treatment (Naslund et al., 2017). This gap is most pronounced in low- and middle-income countries, where resource scarcity, shortage of trained mental health professionals, stigma, and limited infrastructure combine to severely restrict access to care (Alagarajah et al., 2024).

The World Health Organization has identified mental health service accessibility and equity as critical global health priorities. Key barriers to mental health service access include: inadequate numbers of mental health professionals relative to population need, unequal geographic distribution of services (often concentrated in urban centers), high cost of treatment, stigma associated with mental illness, and limited awareness of available services (WHO, 2022). For young adults, the situation is particularly concerning, as this age group experiences high rates of mental health conditions including depression, anxiety, and substance use disorders, yet simultaneously faces unique barriers to accessing care including limited awareness of symptoms, reluctance to seek help due to stigma, and limited knowledge of available services (Alagarajah et al., 2024).

2.4.2 Mental Health Services in Sri Lanka

Sri Lanka, a lower-middle-income country in South Asia, has a developing mental health system characterized by limited specialist resources, inequitable geographic distribution of services, and significant unmet mental health needs among the population (Ministry of Health Sri Lanka, 2023). The National Institute of Mental Health (NIMH) in Colombo serves as the premier psychiatric institution in Sri Lanka and provides inpatient and outpatient psychiatric services; however, availability of mental health services outside of the Western Province remains severely limited (Ministry of Health Sri Lanka, 2023).

Mental health prevalence studies conducted in Sri Lanka have documented significant rates of depression, anxiety, and stress-related disorders across the population, with particular impacts on young adults transitioning through educational and occupational milestones (Perera et al., 2020). The WHO Mental Health Country Profile for Sri Lanka indicates substantial unmet need for mental health services, with treatment-seeking rates remaining low due to stigma, limited awareness, inadequate service availability, and cost barriers (WHO, 2022). Approximately 85-90% of Sri Lankans with mental health conditions do not receive professional treatment (Ministry of Health Sri Lanka, 2023).

The mental health services system in Sri Lanka comprises public psychiatric hospitals and units, private psychiatric clinics, and emerging digital platforms. However, service coverage remains concentrated in urban areas, particularly Colombo, while rural and semi-urban districts face severe shortages of trained psychiatrists, psychologists, and counselors (Ministry of Health Sri Lanka, 2023). The economic burden of mental health conditions on individuals and families is substantial, with cost of treatment representing a significant barrier for lower-income populations (Perera et al., 2020).

2.5 Digital Technologies for Mental Health Service Delivery

Breaks down three key technologies. telepsychiatry (remote consultations), online counseling platforms, and mobile mental health apps for self-management and therapy.

2.5.1 Telepsychiatry and Remote Psychiatric Consultations

Telepsychiatry, the delivery of psychiatric services through video, telephone, or other electronic communication technologies, has emerged as a key strategy for expanding mental health service capacity and improving geographic access to specialist psychiatric care (Torous et al., 2019). Telepsychiatry enables patients to consult with psychiatrists and mental health professionals remotely, eliminating the need for travel to specialized centers and reducing wait times for appointments (Shore et al., 2018). Evidence from multiple jurisdictions demonstrates that telepsychiatry can be delivered effectively and is associated with patient outcomes equivalent to in-person psychiatric care for many conditions (Pierce, 2020).

The feasibility and effectiveness of telepsychiatry in low- and middle-income country contexts has been demonstrated in pilot and implementation studies across South Asia. A study conducted in multiple South Asian countries documented successful telepsychiatry programs connecting rural populations to psychiatrists located in urban centers, with high patient satisfaction and measurable improvements in mental health outcomes (Senanayake & Senanayake, 2017). Telepsychiatry is particularly valuable in Sri Lanka's context, where the concentration of psychiatrists in Colombo and limited availability in provincial areas creates geographic barriers to specialist care (Ministry of Health Sri Lanka, 2023).

2.5.2 Online Counseling and Psychotherapy Platforms

Online counseling and psychotherapy platforms deliver mental health counseling and psychological interventions through text-based chat, video conferencing, or telephone. These platforms expand access to trained counselors and therapists while reducing cost barriers associated with in person counseling (Riva et al., 2016). Online counseling is particularly accessible to young adults, who report preferences for digital communication modalities and express willingness to use online platforms for mental health support (Torous et al., 2017).

Research examining online counseling effectiveness has documented positive outcomes across multiple mental health conditions. A systematic review by Riva et al. (2016) found that online psychological interventions were associated with significant reductions in depression, anxiety, and other mental health symptoms, with effectiveness comparable to face-to-face interventions in many cases. Online counseling platforms have the advantage of providing greater anonymity than in-person services, which may reduce stigma-related barriers to help-seeking, particularly in cultures where mental health stigma is pronounced (Naslund et al., 2017).

2.5.3 Mental Health Applications (Apps) and Self-Management Tools

Mobile mental health applications represent a rapidly expanding category of digital mental health interventions. These applications provide functions including mood and symptom tracking, psychoeducational content, guided self-help interventions, crisis support resources, and peer support features (Torous et al., 2017). Mental health applications range from simple utility tools (e.g., mood tracking, meditation guides) to sophisticated clinical interventions incorporating evidence-based therapeutic approaches such as cognitive behavioral therapy (CBT) and mindfulness-based interventions (Torous et al., 2019).

The World Health Organization has recognized mental health applications as potentially valuable components of digital mental health strategies in resource-limited settings, particularly for preventive interventions and early intervention for individuals experiencing mild to moderate mental health symptoms (WHO, 2019). Studies

conducted in South Asian contexts have documented increasing awareness and willingness to use mental health applications among young adults, though actual usage rates remain variable (Naslund et al., 2017). The accessibility of mental health applications, particularly when offered at low or no cost, represents a significant advantage in low-income populations where cost barriers to traditional mental health services are substantial (Alagarajah et al., 2024).

2.6 Barriers to Digital Mental Health Service Access

Identifies five main barriers. low digital literacy, poor internet infrastructure, cost, privacy and data security concerns, and persistent mental health stigma.

2.6.1 Digital Literacy and Technical Skills

Digital literacy the ability to effectively find, understand, and use information through digital platforms and technologies represents a significant barrier to digital mental health service access, particularly in populations with limited educational background or prior technology exposure (Naslund et al., 2017). Individuals with low digital literacy may find digital mental health platforms difficult to navigate, may struggle to understand technical requirements, and may lack confidence in their ability to use digital health technologies (Aranda-Jan et al., 2014).

In Sri Lankan contexts, digital literacy varies substantially by age, educational level, and geographic location, with rural and older populations demonstrating lower digital literacy than urban and younger populations (Telecommunications Regulatory Commission of Sri Lanka, 2023). While young adults aged 20-34 in urban areas such as Colombo generally have higher levels of digital literacy and smartphone familiarity than older age groups, heterogeneity within this cohort means that some individuals still experience digital literacy barriers (Telecommunications Regulatory Commission of Sri Lanka, 2023). Digital mental health service providers must design platforms and interfaces that accommodate users with varying levels of digital literacy, using clear language, intuitive navigation, and readily available technical support (Torous et al., 2017).

2.6.2 Internet Access and Digital Infrastructure

Access to reliable, affordable internet connectivity represents a critical infrastructure barrier to digital mental health service utilization, particularly in low- and middle-income countries where internet connectivity remains inconsistent and expensive (Naslund et al., 2017). While smartphone penetration has increased substantially in South Asia, including Sri Lanka, reliable broadband internet access for video-based telepsychiatry or streaming high-quality multimedia content remains limited in many regions (Telecommunications Regulatory Commission of Sri Lanka, 2023).

In Sri Lanka, mobile internet penetration has increased significantly in urban areas, with mobile telecommunications serving as the primary means of internet access for most users; however, connectivity remains intermittent in rural and semi-rural areas (Telecommunications Regulatory Commission of Sri Lanka, 2023). The cost of mobile internet data in Sri Lanka represents a barrier for lower-income users, as continuous use of video-based telepsychiatry or frequent access to data-intensive applications can accumulate substantial costs (Perera et al., 2020). Digital mental health services designed for low-resource contexts must be compatible with limited bandwidth and low-data environments, utilizing text-based or simple graphical interfaces rather than demanding high-bandwidth video or multimedia content (Naslund et al., 2017).

2.6.3 Cost Barriers

Cost represents both a direct and indirect barrier to digital mental health service access. Direct costs include device costs (smartphones or computers), internet data costs, and fees charged for accessing digital mental health platforms or services. Indirect costs include time and transportation costs associated with seeking digital mental health services (Naslund et al., 2017). Even in lower-resource contexts, digital health services perceived as more affordable than traditional in-person care can still represent significant costs for lower-income individuals or families (Aranda Jan et al., 2014).

In Sri Lanka, where mean household income levels are lower than in developed nations, cost barriers to digital mental health services are substantial. Although smartphones have become more affordable and mobile internet rates have declined, many individuals still face financial constraints that limit their ability to maintain continuous internet access or to pay for premium mental health applications or services (Telecommunications Regulatory Commission of Sri Lanka, 2023). Successful digital mental health strategies in Sri Lanka must incorporate free or substantially subsidized access to digital mental health services, with funding support from government health budgets or non-governmental organizations (Alagarajah et al., 2024).

2.6.4 Privacy, Confidentiality, and Data Security Concerns

Privacy concerns related to digital mental health platforms represent a significant barrier, particularly in cultural contexts where mental health stigma is pronounced and individuals fear that disclosure of mental health information could result in social or occupational discrimination (Perera et al., 2020). Users of digital mental health platforms express concerns about data security, privacy of stored health information, and potential unauthorized access to sensitive mental health information (Torous et al., 2017). These privacy concerns are particularly acute in developing-country contexts where data protection regulations may be nascent and users have limited recourse if data breaches occur (Naslund et al., 2017).

In Sri Lanka, where mental health stigma remains significant and privacy concerns are culturally important, the assurance of confidentiality and security of digital mental health services is essential to encourage utilization (Perera et al., 2020). Digital mental health platform developers and service providers must implement robust data security measures, obtain explicit informed consent for data collection and use, and provide transparent information about data handling practices (Torous et al., 2017).

2.6.5 Mental Health Stigma and Social Attitudes

Mental health stigma negative attitudes, stereotypes, and discrimination directed at individuals with mental health conditions remains a significant barrier to mental health service-seeking in Sri Lanka and across South Asia (Perera et al., 2020). While digital mental health services offer greater anonymity than in-person services and may reduce some stigma-related barriers, stigma associated with mental health conditions and treatment-seeking persists and may continue to inhibit utilization even of digital platforms (Naslund et al., 2017). Social norms that discourage help-seeking for mental health problems, family attitudes that prioritize concealment of mental health difficulties, and concerns about being judged or socially excluded continue to create barriers to mental health service access (Perera et al., 2020).

Addressing stigma-related barriers to digital mental health service access requires not only technology development but also complementary mental health awareness and anti-stigma campaigns that promote understanding of mental health conditions and normalize treatment-seeking (WHO, 2022). Digital mental health platforms can incorporate peer support features and educational content that contribute to stigma reduction, positioning mental health conditions as treatable and recovery as possible (Torous et al., 2017).

2.7 Enabling Factors for Digital Mental Health Service Adoption

Highlights what helps adoption. high mobile penetration, government policy support, mental health awareness campaigns, social support networks, and availability of multilingual/culturally adapted platforms.

2.7.1 Mobile Penetration and Digital Infrastructure Development

High rates of mobile phone ownership, particularly smartphone ownership, constitute an enabling factor for digital mental health adoption. In Sri Lanka, mobile penetration has reached approximately 120% (more than one mobile device per capita), with smartphone ownership highest among younger age groups (Telecommunications Regulatory Commission of Sri Lanka, 2023). The widespread availability of

smartphones creates the infrastructure foundation necessary for mobile-based mental health applications and services (Naslund et al., 2017).

Digital infrastructure development, including expansion of mobile broadband networks and improvement of internet quality and affordability, progressively enables digital health service access (Telecommunications Regulatory Commission of Sri Lanka, 2023). Government investment in digital infrastructure, telecommunications liberalization, and competition among service providers can reduce internet costs and improve connectivity, particularly in underserved areas, thereby enabling greater digital health access (Alagarajah et al., 2024).

2.7.2 Government Policy Support and Strategic Planning

Government commitment to digital health development, reflected in national digital health strategies, policy frameworks, and funding allocations, represents a critical enabling factor for digital mental health implementation (WHO, 2019). Countries that have incorporated digital health into national health strategies and allocated dedicated funding for digital health infrastructure, training, and service provision have experienced more rapid and sustainable digital health expansion (Naslund et al., 2017).

In Sri Lanka, government recognition of mental health as a health priority and inclusion of digital technologies in health service expansion strategies could facilitate more rapid adoption and integration of digital mental health services into the public health system (Ministry of Health Sri Lanka, 2023). Policy support for telemedicine licensing, reimbursement mechanisms for digital mental health services, and regulatory frameworks that ensure quality and safety of digital mental health platforms are essential for sustainable expansion of digital mental health services (WHO, 2019).

2.7.3 Awareness Campaigns and Health Literacy

Mental health awareness campaigns and initiatives that increase community understanding of mental health conditions and available services constitute important enabling factors for digital mental health adoption (WHO, 2022). Awareness campaigns that specifically promote digital mental health options and demystify technology-based service delivery can increase willingness to utilize digital platforms

(Torous et al., 2017). Health literacy knowledge and understanding of health conditions and health services predicts greater engagement with health services, including digital mental health services (Naslund et al., 2017).

Mental health awareness initiatives in Sri Lanka that incorporate digital mental health information and promote understanding of available digital options could increase service utilization and reduce stigma-related barriers. Integration of mental health education into school curricula, workplace health programs, and community settings can contribute to improved mental health literacy and awareness of digital mental health options (WHO, 2022).

2.7.4 Social Support Networks

Social support the perceived or actual availability of assistance and encouragement from family, friends, and community members represents an important enabling factor for mental health service-seeking and sustained engagement with mental health services, including digital platforms (Perera et al., 2020). Individuals with supportive family relationships and strong social networks are more likely to seek mental health help and to persist in treatment (Perera et al., 2020). Social support can also facilitate digital technology adoption, with peers and family members providing encouragement, technical assistance, and normalization of digital health service use (Venkatesh et al., 2003).

In Sri Lankan cultural contexts, family influence and social networks exert strong influence on health-seeking behavior and technology adoption. Digital mental health initiatives that engage family members and community leaders in promoting understanding and acceptance of digital mental health services can leverage social support as an enabling factor (Perera et al., 202

2.7.5 Multilingual Platform Availability and Cultural Adaptation

Digital mental health platforms designed specifically for diverse linguistic and cultural contexts represent an important enabling factor. Availability of digital mental health services in local languages (Sinhala, Tamil, English) and adaptation of content to reflect cultural values, beliefs, and communication preferences increases accessibility and

acceptability (Naslund et al., 2017). Technology designed with attention to cultural appropriateness, local values, and community-specific mental health concerns is more likely to be adopted and utilized (Alagarajah et al., 2024).

In Sri Lanka, digital mental health platforms that offer content in Sinhala and Tamil, that reflect Sri Lankan cultural contexts, and that incorporate locally relevant mental health resources would enhance acceptability and utilization compared to generic international platforms (Ministry of Health Sri Lanka, 2023).

2.8 Young Adults and Mental Health: Characteristics and Service Access

Focuses on why young adults (18-35) are particularly vulnerable to mental health conditions and why they are also the most likely to prefer and use digital mental health services.

2.8.1 Mental Health Burden Among Young Adults

Young adults (typically defined as ages 20-34) experience high prevalence rates of mental health conditions, particularly depression, anxiety disorders, and stress-related conditions (WHO, 2022). This life stage involves multiple transitions including educational completion, entry into the workforce, establishment of independent living situations, relationship formation, and decisions about family planning all of which create psychological stressors and increase vulnerability to mental health conditions (Alagarajah et al., 2024). The World Health Organization identifies young adulthood as a critical period for mental health, noting that the majority of mental health conditions emerge by the late teens or early twenties (WHO, 2022).

In South Asian populations, including Sri Lanka, young adults report substantial rates of depression, anxiety, and stress-related symptoms, often related to educational pressures, employment challenges, family expectations, and social transitions (Perera et al., 2020). Mental health conditions in young adulthood, if left untreated, are associated with long-term impacts on educational attainment, occupational functioning, relationship quality, and physical health (WHO, 2022). Early intervention during young adulthood represents an opportunity to prevent progression to more severe mental

health conditions and to support optimal functioning across life domains (Alagarajah et al., 2024).

2.8.2 Young Adults' Preferences for Digital Mental Health Services

Young adults, as a digitally native generation with high rates of smartphone ownership, internet access, and prior experience with digital communication technologies, demonstrate preferences for digital mental health service delivery modalities (Torous et al., 2017). Research examining young adults' attitudes toward digital mental health has documented high willingness to use mental health applications, online counseling, and telepsychiatry, with cited advantages including convenience, reduced stigma, ability to access services at flexible times, and greater perceived anonymity (Naslund et al., 2017).

Studies conducted in South Asian contexts, including Sri Lanka, have documented similar patterns among young adults, with high ownership of smartphones, regular use of social media and digital communication, and expressed openness to digital mental health services (Telecommunications Regulatory Commission of Sri Lanka, 2023). Young adults in Sri Lanka indicate willingness to use mental health applications and online counseling services, particularly when services are free or low-cost and when privacy and confidentiality are assured (Perera et al., 2020).

2.8.3 Barriers to Mental Health Service Seeking in Young Adults

Despite high mental health burden and expressed willingness to use digital services, young adults experience specific barriers to mental health service access. These include: limited mental health literacy and awareness of available services, stigma and concerns about others' judgment, preference for self-reliance and reluctance to disclose personal difficulties, limited financial resources in many cases, and low perceived need or minimization of symptoms (Alagarajah et al., 2024). Young adults may lack awareness of mental health conditions (not recognizing symptoms as indicating a mental health problem), may perceive symptoms as temporary or situational rather than requiring professional intervention, and may lack knowledge of available services (Torous et al., 2017).

In Sri Lankan contexts, young adults' mental health service-seeking is further inhibited by stigma associated with mental health treatment, family concerns about confidentiality, limited availability of age-appropriate services, and cost barriers (Perera et al., 2020). Digital mental health services that address these barriers through easy accessibility, improved privacy, affordability, and age-appropriate design have potential to significantly improve mental health service access among this population (Alagarajah et al., 2024).

2.9 Digital Technology and Access to Mental Health Services

Conceptualizes "access" across four dimensions: availability, accessibility, acceptability, and affordability and argues that digital technology can address all four.

2.9.1 Conceptualizing Access to Mental Health Services

Access to mental health services encompasses multiple dimensions: availability (whether services exist in a given geographic area), accessibility (ease of reaching services given distance, transportation, and time constraints), acceptability (whether services match population expectations and preferences), and affordability (whether services are financially feasible for intended users) (Penchansky & Thomas, 1981). Beyond these structural dimensions, access also includes perceptual dimensions such as perceived ease of access, awareness of available services, and perceived likelihood of receiving needed care (Naslund et al., 2017).

Access to mental health services in Sri Lanka is constrained across multiple dimensions: limited geographic availability of services outside Colombo (structural availability barrier), high cost of private psychiatric care (affordability barrier), lack of awareness of available services among population (perceptual barrier), and limited number of trained mental health professionals relative to population need (capacity barrier) (Ministry of Health Sri Lanka, 2023).

2.9.2 Digital Technology as a Strategy for Improving Mental Health Service Access

Digital technologies can address multiple dimensions of mental health service access constraints. Telepsychiatry and online counseling services can overcome geographic

barriers by enabling individuals in remote areas to access specialists located in urban centers without requiring travel (Shore et al., 2018). Mobile mental health applications can provide mental health information, psychoeducation, and self-management tools to individuals who might not otherwise access any mental health services, expanding the reach of mental health support beyond formal healthcare systems (Torous et al., 2019). Digital platforms can operate at reduced cost compared to in-person services, potentially improving affordability (Naslund et al., 2017).

Digital mental health services provide 24/7 availability compared to time-limited clinic hours, improving temporal access for individuals with scheduling constraints (Torous et al., 2017). Online platforms can provide greater anonymity and perceived privacy compared to in-person services, potentially reducing stigma-related barriers to service-seeking, particularly in cultural contexts where mental health stigma is pronounced (Perera et al., 2020).

The potential of digital technologies to improve mental health service access is greatest when digital implementations are integrated into comprehensive mental health strategies that also address non-digital barriers such as stigma, low mental health literacy, and lack of awareness (Naslund et al., 2017). Evidence from multiple jurisdictions demonstrates that digital mental health technologies are most effective when supported by complementary stigma reduction initiatives, mental health awareness campaigns, and integration with existing mental health service infrastructure (Alagarajah et al., 2024).

2.10 Empirical Evidence on Digital Mental Health Effectiveness

Reviews research showing that telepsychiatry, online counseling (dCBT), and mental health apps all produce outcomes comparable to face-to-face care, though app dropout rates remain a challenge.

2.10.1 Effectiveness of Telepsychiatry and Remote Mental Health Consultations

Systematic reviews and meta-analyses examining telepsychiatry effectiveness have documented outcomes comparable to in-person psychiatric care across multiple mental health conditions including depression, anxiety, bipolar disorder, and psychotic

disorders (Pierce, 2020). A meta-analysis of telepsychiatry studies found effect sizes for symptom reduction equivalent to face-to-face care, with patient satisfaction with telepsychiatry services generally high and dropout rates similar to in-person care (Shore et al., 2018).

Implementation studies of telepsychiatry in developing-country contexts have demonstrated feasibility and effectiveness. A study of telepsychiatry programs in South Asia found that remote psychiatric consultations were associated with significant symptom improvements among individuals with depression and anxiety, with high patient satisfaction and cost savings compared to in-person specialist care (Senanayake & Senanayake, 2017).

2.10.2 Effectiveness of Online Counseling and Digital Psychotherapy

Research on online counseling and digital psychotherapy interventions, including dCBT and computerized mental health interventions, has documented significant effectiveness. A systematic review and meta-analysis found that online psychological interventions were associated with significant reductions in depression and anxiety symptoms, with effect sizes comparable to face-to-face interventions (Riva et al., 2016). Online counseling platforms have demonstrated particular effectiveness for mild to moderate depression and anxiety, though utilization for severe mental health conditions may require supplementation with in-person care (Torous et al., 2019).

2.10.3 Effectiveness of Mobile Mental Health Applications

Mental health mobile applications incorporating evidence-based intervention components have demonstrated effectiveness for symptom monitoring, psychoeducation, and self-management. Studies of mobile applications incorporating cognitive behavioral therapy modules, mindfulness-based interventions, and mood tracking have documented user satisfaction and improvements in mental health symptoms (Torous et al., 2017). However, high dropout rates and inconsistent engagement with mental health applications represent a significant challenge, with many users downloading applications but utilizing them for limited periods (Torous et al., 2019).

2.11 Digital Divide and Health Inequities

Warns that digital mental health could worsen existing inequalities if it only reaches those already digitally advantaged, and calls for equity-focused design strategies.

2.11.1 The Digital Divide

The digital divide refers to disparities in access to digital technologies, internet connectivity, and digital literacy across different populations, often reflecting and reinforcing existing socioeconomic inequalities (Norris, 2001). First-level digital divide encompasses disparities in physical access to digital devices and internet connectivity, while second-level digital divide encompasses disparities in digital skills and ability to effectively utilize digital technologies (Helsper & van Deursen, 2016).

In Sri Lankan contexts, the digital divide exists along multiple dimensions: geographic (urban-rural differences in internet infrastructure), socioeconomic (lower-income populations having less access to devices and connectivity), educational (lower-education populations having lower digital literacy), and demographic (age related disparities in technology adoption) (Telecommunications Regulatory Commission of Sri Lanka, 2023). The digital divide has direct implications for digital health equity; digital mental health services may inadvertently deepen health inequities by benefiting digitally advantaged populations while excluding those without digital access or literacy (Naslund et al., 2017).

2.11.2 Ensuring Digital Mental Health Equity

Strategies for ensuring digital mental health equity include: designing digital platforms accessible to users with limited digital literacy, developing services compatible with low-bandwidth and limited-data environments, addressing cost barriers through free or subsidized services, ensuring linguistic and cultural appropriateness, and integrating digital mental health services with non-digital mental health service options to ensure no population is left behind (Naslund et al., 2017). Particular attention to reaching marginalized and vulnerable populations including lower-income individuals, rural populations, and groups experiencing multiple forms of discrimination is essential to

ensure digital mental health benefits all segments of the population (Alagarajah et al., 2024).

2.12 Chapter Summary

These findings justify investigation of the effectiveness of digital technology in improving access to mental health services among young adults in Colombo, Sri Lanka, as an initial empirical assessment in an urban, digitally-active context that can inform future expansion to other regions. Understanding the mechanisms by which digital technology adoption, barriers, and enabling factors determine access to mental health services and the role of digital health engagement in mediating these relationships will provide evidence to support evidence-based digital mental health policy and implementation in Sri Lanka and comparable low-middle-income country contexts.

Chapter Three: Research Methodology

3.1 Introduction

This chapter delineates the research methodology adopted to investigate Digital Technology as a Catalyst for Improving Access to Mental Health Services among Young Adults in Sri Lanka, with a specific focus on Colombo. The research employs a quantitative approach utilizing primary data collection methods to examine how digital health engagement mediates the relationship between digital technology implementation and access to mental health services among young adults aged 20-34 years.

3.2 Research Method

A quantitative research design was selected as the most epistemologically and methodologically appropriate approach for achieving the study's objectives. Quantitative designs are well suited to studies that seek to measure variables precisely, test theoretically derived hypotheses, identify and quantify relationships among constructs, and generate findings that can be generalized to a broader population within the confidence limits of the sample (Creswell & Creswell, 2018). Given that the central aim of this study is to evaluate the effectiveness of digital technology in improving access to mental health services a research question that requires the measurement of adoption rates, perceptual attitudes, barriers, and enabling factors across a defined population segment the quantitative paradigm is the appropriate choice.

Within the broad umbrella of quantitative research, a cross-sectional survey design is employed. Cross-sectional designs involve the collection of data from a representative sample of the target population at a single point in time, allowing the researcher to describe the current state of variables of interest and to examine the associations and causal relationships among those variables (Babbie, 2020). This design is both cost-effective and practically feasible for a student dissertation undertaken within the University of Kelaniya, and is widely used in digital health research in South and Southeast Asia (Naslund et al., 2017).

The study is explanatory in orientation, seeking not merely to describe patterns of digital technology use among young adults in Colombo, but to explain through the

testing of empirically grounded hypotheses the mechanisms by which digital technology adoption, barriers, and enabling factors determine the level of access to mental health services. This explanatory character is entirely consistent with the application of Structural Equation Modeling (SEM) as the principal analytical technique for the main objective, and with the use of Chi-Square tests of association and goodness-of-fit for the sub-objectives. The study aligns with a positivist epistemological stance, positing that social and health-related phenomena are measurable through systematic, objective methods and that knowledge is derived through the empirical testing of hypotheses (Bryman, 2016). The research utilizes primary data collection exclusively. This research employs an online questionnaire administered via Google Forms as the primary data collection instrument. Primary data collection ensures the relevance and accuracy of information gathered relative to the specific research objectives and variables being measured.

3.3 Study Area and Population

3.3.1 Study Area

The study is conducted within the Colombo District, the administrative, commercial, and digital hub of Sri Lanka. Colombo is the most densely populated district in the country and hosts the highest concentration of internet service providers, mobile network infrastructure, healthcare institutions, and technology enterprises, making it the most suitable setting for studying digital mental health behaviors among urban young adults (Telecommunications Regulatory Commission of Sri Lanka, 2023). From a mental health services perspective, Colombo contains the National Institute of Mental Health (NIMH), the largest psychiatric hospital in Sri Lanka, alongside numerous public and private counselling centers, outpatient psychiatric units, and emerging telepsychiatry and digital wellness platforms (Ministry of Health Sri Lanka, 2023). The district therefore provides both the digital infrastructure and the healthcare ecosystem necessary to investigate the intersection of digital technology and mental health service access.

While the urban character of Colombo limits the direct generalizability of findings to rural or semi-urban districts, it offers the most favorable conditions for digital

technology adoption and is therefore the most appropriate setting for an initial empirical assessment. Findings from Colombo can serve as a reference point and benchmark for future comparative studies in other districts of Sri Lanka (Senanayake et al., 2020). The study area encompasses all administrative divisions within the Colombo District as delineated by the Department of Census and Statistics.

3.3.2. Population of the study

The target population comprises non-clinical young adults between the ages of 20 and 34 who are residing in the Colombo District. This age cohort is selected because it represents the most digitally active segment of the Sri Lankan population, exhibiting the highest rates of smartphone ownership and internet use (Telecommunications Regulatory Commission of Sri Lanka, 2023), while simultaneously being disproportionately affected by common mental health conditions such as depression, anxiety, and stress-related disorders (WHO, 2022). Moreover, young adults in the 20-34 age range constitute a critical life stage during which mental health interventions including digitally delivered ones can yield the greatest long-term benefits in terms of productivity, social functioning, and prevention of severe outcomes such as suicidality and substance misuse (Alagarajah et al., 2024).

The target population is further defined to include only non-clinical individuals that is, those without a formal diagnosis of a severe or chronic mental health disorder at the time of the study. This delimitation is intentional, as the study focuses on preventive and early-intervention dimensions of digital mental health, which are considered the most scalable and cost-effective approaches in resource-limited settings such as Sri Lanka (Naslund et al., 2017).

The population (N) for the purpose of sample size calculation is operationalized using the most authoritative available source of district-level, age-disaggregated population data: the Census of Population and Housing 2012 conducted by the Department of Census and Statistics of Sri Lanka. The 2012 Census remains the definitive demographic reference for Sri Lanka at the district level, as a subsequent national census has yet to be completed. The relevant population figures for Colombo District across the three targeted age strata are presented in Table 3.1.

Table 3.1: Study Population by Age- Group (Colombo District)

AGE GROUP	TOTAL (N _i)
20 – 24 YEARS	200,897
25 – 29 YEARS	167,913
30 – 34 YEARS	158,469
TOTAL	527,279

Source: Department of Census and Statistics, Sri Lanka (2024)

As shown in Table 3.1, the combined population of the three targeted age strata amounts to approximately 527,279. This figure derived by summing the sub-totals for the 20–24 years stratum (200,897), the 25–29 years stratum (167,913), and the 30–34 years stratum (158,469) is accepted as the study population (N) for all subsequent sample size calculations. It is acknowledged that some demographic change may have occurred since 2024; however, in the absence of a more recent census, the 2024 data constitute the most scientifically defensible basis for population estimation (Department of Census and Statistics, 2024).

3.4 Sample and Sampling Process

Given the large size of the study population ($N = 527,279$) and the need for a statistically defensible yet practically achievable sample size, the Yamane (1967) simplified random sampling formula is applied. This formula is widely used in social science research involving large finite populations and has been applied in numerous digital health studies conducted in South Asian settings (Senanayake & Senanayake, 2017; Adikari et al., 2022). The formula is expressed as:

$$n = N / (1 + N(e)^2)$$

A margin of error of $e = 0.1$ (10%) is adopted, reflecting a pragmatic balance between statistical precision and the resource constraints of a student dissertation conducted within a defined geographic area. A 10% margin of error is a widely accepted standard in social science research in developing country contexts, particularly for exploratory and explanatory studies that aim to identify directional relationships among variables

rather than produce precise population-level prevalence estimates (Israel, 1992; Yamane, 1967). Substituting the known values into the formula:

$$n = 527,279 / (1 + 527,279 \times (0.10)^2)$$

$$n = 527,279 / (1 + 527,279 \times 0.01)$$

$$n = 527,279 / (1 + 5,272.79)$$

$$\mathbf{n = 527,279 / 5,273.79 \approx 99.98 \approx 100}$$

The required sample size is therefore $n \approx 100$. This value is sufficient to yield statistically meaningful results at the 90% confidence level and is consistent with sample sizes employed in comparable digital health studies in Sri Lanka and other South Asian countries (Adikari et al., 2022). The computed sample size of 100 participants is used as the basis for all subsequent sampling allocation and data collection procedures.

3.4.1. Sampling Technique

To ensure that each of the three age strata within the target population is proportionately and adequately represented in the sample, a proportionate stratified random sampling technique is employed. Stratified random sampling involves dividing the study population into mutually exclusive and collectively exhaustive subgroups called strata defined on the basis of a relevant characteristic, and then drawing a random sample from each stratum (Cochran, 1977). This approach is superior to simple random sampling when the study population is heterogeneous with respect to a key classification variable in this case, age and when the researcher requires that all subgroups are represented in the sample in proportion to their size in the population (Babbie, 2020).

Three strata are defined based on the age groups of the target population: Stratum I (20–24 years), Stratum II (25–29 years), and Stratum III (30–34 years). The sample is allocated to each stratum in proportion to that stratum's share of the total study population, using the proportionate allocation formula:

$$n_i = (N_i / N) \times n$$

Applying this formula to each stratum yields the sample allocation shown in Table 3.2. Within each stratum, individual respondents are selected through simple random sampling, implemented through snowball distribution of the Google Form survey link across multiple online channels including university email lists, professional social media groups (LinkedIn, Facebook) and WhatsApp community networks operating within Colombo. This digital recruitment strategy is contextually appropriate given the digitally active nature of the target population and the digital focus of the research.

Table 3.2: Proportionate Stratified Sample Allocation

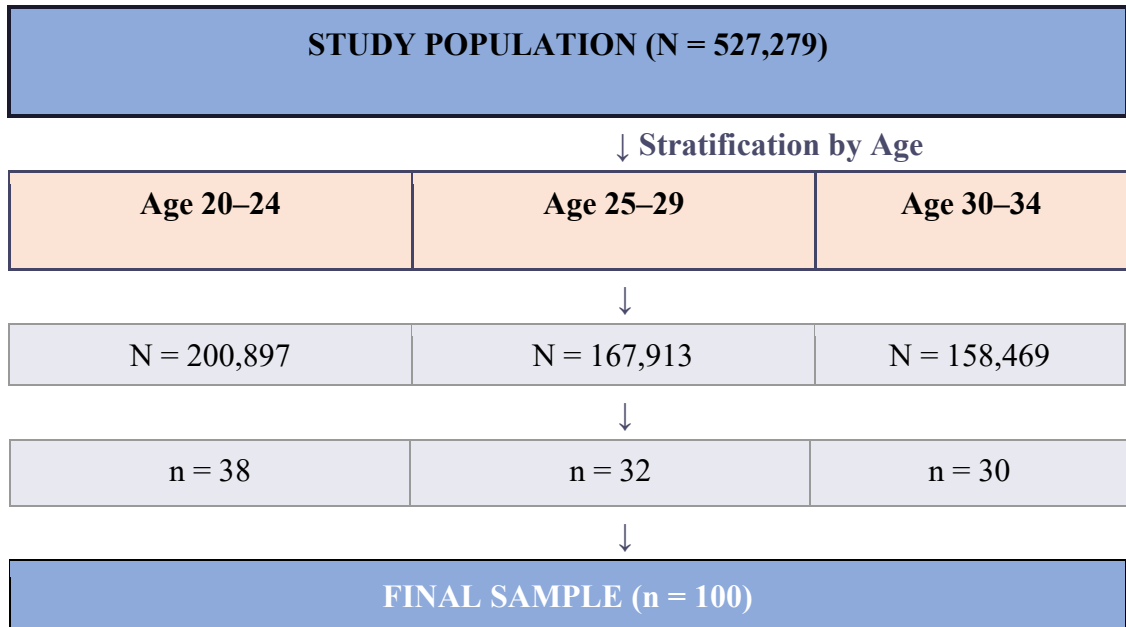
Stratum	Age Group	Population (N _i)	Proportion	Sample (n _i)
I	20 – 24 Years	200,897	37.9%	38
II	25 – 29 Years	167,913	31.6%	32
III	30 – 34 Years	158,469	30.5%	30
Total				100

Source: Compiled by researcher using Department of Census and Statistics (2024) and Yamane (1967)

As demonstrated in Table 3.2, the proportionate allocation yields 34 respondents from Stratum I, 33 from Stratum II, and 33 from Stratum III, summing to a total sample of $n = 100$. Minor rounding adjustments are applied to ensure integer values while maintaining the overall sample size. This allocation ensures that the relative demographic composition of the sample faithfully mirrors that of the target population in the Colombo District.

The complete sampling procedure, from the identification of the study population through to the final allocation of respondents across the three age strata, is illustrated in Figure 3.1.

Figure 3.1: Stratified Random Sampling Procedure



Source: Department of Census and Statistics Sri Lanka (2012).

The sampling procedure was operationalized through the following steps are listing all identifiable young adults within the relevant age groups in targeted community locations across Colombo, including universities, workplaces, and community centers, stratifying the list by age group, applying proportionate random selection within each stratum using a random number table and administering the questionnaire to selected respondents after obtaining informed consent.

3.5 Data Collection Method

This study relies exclusively on primary data-data collected directly from respondents by the researcher for the specific and original purpose of answering the research questions of the present investigation (Saunders, Lewis, & Thornhill, 2019). Primary data collection is the appropriate choice because no existing secondary dataset captures the specific combination of constructs under investigation digital technology adoption, barriers, enabling factors, and perceived access to mental health services among the defined target population in the Colombo District. The use of primary data ensures that the collected information is both current and directly tailored to the study's analytical

framework, making it amenable to the SEM and Chi-Square tests prescribed by the research design.

The primary data collection instrument is a structured, self-administered questionnaire deployed digitally via Google Forms. Google Forms is selected as the delivery platform for several methodologically and contextually justified reasons. First, the study's target population of young adults in Colombo is characterized by relatively high rates of smartphone ownership and internet connectivity (Telecommunications Regulatory Commission of Sri Lanka, 2023), making online delivery both accessible and contextually consistent with the subject matter of the research. Second, the anonymous and self-administered nature of the online instrument is particularly advantageous given the sensitive nature of mental health topics in the Sri Lankan cultural context, where stigma may inhibit honest responses in face-to-face or interviewer-administered settings (Senanayake et al., 2020).

3.6 Reliability and Consistency Testing

Reliability refers to the degree to which a measurement scale produces stable, consistent results across repeated administrations and across the items that are intended to measure the same underlying construct (Nunnally & Bernstein, 1994). In quantitative research employing multi-item Likert scales, it is essential to verify the internal consistency of each scale before proceeding to inferential analysis. This is particularly critical in SEM, which assumes that latent constructs are reliably operationalized by their observed indicators; failure to confirm adequate reliability can lead to the underestimation of structural path coefficients and to spurious or misleading conclusions (Hair et al., 2014). Equally, for the Chi-Square analyses applied to the sub-objectives, scale reliability is important to ensure that the composite or grouped variables used in the cross-tabulations accurately capture the underlying constructs they purport to represent. The standard index of internal consistency reliability employed in this study is Cronbach's Alpha (α), introduced by Cronbach (1951) and subsequently endorsed by numerous methodologists as the primary measure of scale reliability in

social science research (George & Mallery, 2003; Field, 2018). Cronbach's Alpha ranges from 0 to 1, with higher values indicating greater inter-item consistency. The

following interpretive framework is adopted, consistent with George and Mallery (2003): $\alpha < 0.60$ = poor; $0.60 \leq \alpha < 0.70$ = questionable; $0.70 \leq \alpha < 0.80$ = acceptable; $0.80 \leq \alpha < 0.90$ = good; $\alpha \geq 0.90$ = excellent. A minimum threshold of $\alpha \geq 0.70$ is applied in this study, consistent with the recommendation of Nunnally and Bernstein (1994) and widely applied in digital health research in developing country contexts (Naslund et al., 2017).

In addition to the overall Alpha value, item-total correlations will be examined for each scale; items with corrected item-total correlations below 0.30 will be considered for removal from the scale to improve internal consistency (Field, 2018). The Alpha if item deleted statistic will also be inspected to identify any item whose removal would substantially improve the overall Alpha coefficient. Table 3.3 presents the four measurement scales, the number of items in each, and the minimum expected Cronbach's Alpha coefficients based on the pilot test and the existing literature on similar constructs.

Table 3.3: Internal Consistency Reliability

Scale / Construct	Min. Acceptable α	Reliability Level
Digital Technology Adoption (DTA)	≥ 0.70	Acceptable
Barriers (BAR)	≥ 0.70	Acceptable
Enabling Factors (ENF)	≥ 0.70	Acceptable
Access to Mental Health Services (AMHS)	≥ 0.70	Acceptable

Source: Compiled by researcher based on Cronbach (1951), George & Mallery (2003), and Nunnally & Bernstein (1994)

In addition to Cronbach's Alpha, the study will assess composite reliability ($CR \geq 0.70$) and average variance extracted ($AVE \geq 0.50$) within the SEM framework to evaluate

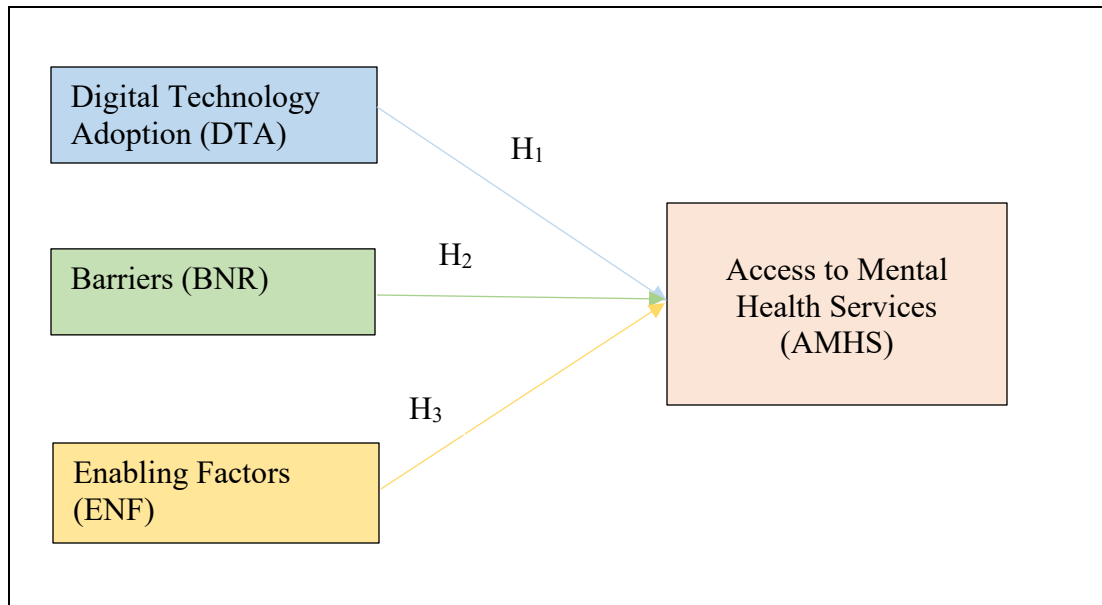
convergent validity. Discriminant validity will be assessed using the Fornell Larcker (1981) criterion, which requires that the square root of each construct's AVE exceeds its correlation with all other constructs. Together, these assessments ensure that the measurement model is sufficiently robust before structural relationships are estimated.

3.7 Conceptual Framework

The research framework, developed from the theoretical and empirical literature reviewed in Chapter Two, provides a structured conceptual map of the causal relationships among the key constructs investigated in this study. The framework posits that access to mental health services among young adults in Colombo the study's dependent variable is determined by three independent constructs: Digital Technology Adoption (DTA), Barriers (BAR), and Enabling Factors (ENF). This structure is consistent with the Digital Divide framework (Norris, 2001), and the body of empirical evidence on digital mental health in low- and middle-income countries (Naslund et al., 2017; Alagarajah et al., 2024).

Figure 3.2 presents a complete specification of the study's variables, their classifications as independent or dependent, and the observable indicator items used to measure each construct in the questionnaire.

Figure 3.2: Conceptual Framework



Source: Compiled by researcher from the theoretical framework and review of literature

The conceptual framework of this study comprises three independent variables, namely Digital Technology Adoption (DTA), Barriers (BNR), and Enabling Factors (ENF), and one dependent variable, Access to Mental Health Services (AMHS). These relationships are examined simultaneously through Structural Equation Modelling (SEM) with path analysis, where each independent variable is hypothesized to exert a distinct directional influence on the dependent variable among young adults in Colombo, Sri Lanka.

The hypothesis H₁ posits that DTA has a significant positive influence on AMHS. As individuals become more proficient and comfortable in using digital tools such as mobile mental health applications, telepsychiatry platforms, and online counselling services, their ability to access mental health support is expected to increase considerably. Therefore, it is hypothesized that young adults who demonstrate higher levels of digital technology adoption will report significantly greater access to mental health services through digital platforms.

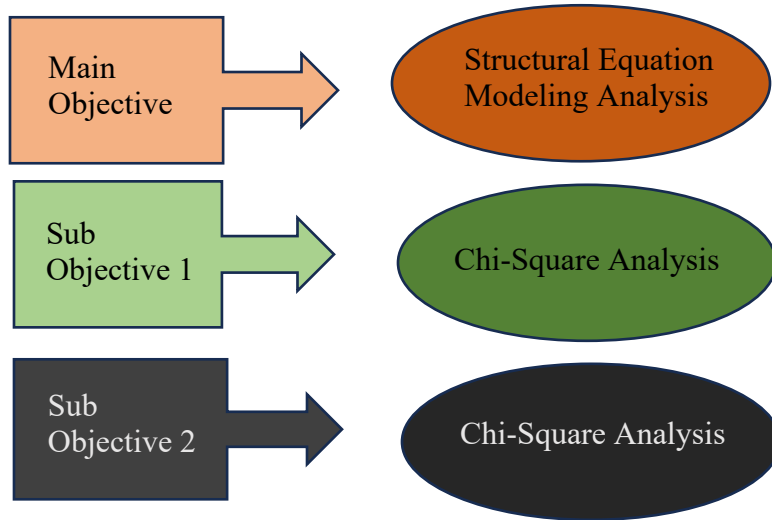
The hypothesis H₂ proposes that BNR have a significant negative influence on AMHS. Barriers encompass a range of structural, cultural, and socioeconomic obstacles, including limited digital literacy, financial constraints, social stigma associated with mental health help-seeking, and inadequate technological infrastructure. These factors are expected to impede young adults from effectively utilizing digital mental health services, thereby reducing their overall level of access. It is therefore hypothesized that higher levels of perceived barriers will be significantly associated with lower levels of access to mental health services through digital technology.

The third hypothesis H₃ suggests that ENF have a significant positive influence on AMHS. Enabling factors refer to the supportive conditions that facilitate the adoption and use of digital mental health services, including the availability of affordable smartphones, strong internet connectivity, government health initiatives, social media awareness campaigns, and peer support networks. When these enabling conditions are present and accessible, young adults are better positioned to engage with and benefit from digital mental health platforms. Accordingly, it is hypothesized that higher levels of enabling factors will be significantly associated with greater access to mental health services among the target population in Colombo, Sri Lanka.

3.8 Data Analysis Techniques

After data collection and initial cleaning in all data were imported descriptive statistics, reliability tests, and Chi-Square analyses, and SEM analysis. The choice of specific analytical techniques was guided by the level of measurement of the variables, the nature of the hypotheses, and the information needs of each research objective. Figure 3.3 presents the analytical techniques applied for each objective.

Figure 3.3: Data Analysis Techniques for Objectives



Source: Compiled by researcher

3.8.1 Structural Equation Modeling

SEM is employed as the principal analytical technique for the main research objective to evaluate the effectiveness of digital technology in improving access to mental health services. SEM is selected because it uniquely integrates a measurement model (Confirmatory Factor Analysis, CFA) and a structural model (path analysis) within a single unified framework, enabling the simultaneous estimation of direct and indirect causal pathways among multiple latent constructs while accounting for measurement error in each observed indicator (Kline, 2016; Hair et al., 2014). This methodological advantage is critical in the present study, where the constructs of interest DTA, BAR, ENF, and AMHS are theoretical latent variables that cannot be directly observed but must be inferred from their observable indicators.

The SEM analysis is executed in IBM AMOS (Version 29) using Maximum Likelihood Estimation (MLE), which is the most robust and widely used estimation method for SEM with continuous or approximately continuous Likert-scale indicators (Hair et al., 2014). The SEM proceeds in two sequential stages. In Stage 1, a Confirmatory Factor Analysis (CFA) is conducted to test the fit of the measurement model verifying that the Likert-scale items in Sections 1, 2, and 3 load significantly onto their respective latent constructs and that the model demonstrates acceptable convergent and discriminant

validity (Fornell & Larcker, 1981). In Stage 2, conditional on acceptable measurement model fit, the structural model is estimated with the hypothesized causal paths specified. The significance of structural path coefficients (β) is assessed using critical ratio (CR) values, with $|CR| \geq 1.96$ corresponding to $p < 0.05$ (Kline, 2016).

Indirect effects particularly the mediated path from ENF to AMHS via DTA will be estimated using bootstrapping with 5,000 resamples and bias-corrected 95% confidence intervals, as recommended by Preacher and Hayes (2008) to avoid the distributional assumptions of the Sobel test. All SEM results will be reported with standardized path coefficients, p-values, and overall model fit statistics.

3.8.2 Chi-Square Analysis

Chi-Square (χ^2) statistical tests are employed as the primary analytical tools for addressing both sub-objectives of this study. The Chi-Square test is a non-parametric statistical technique that examines whether there is a statistically significant association between two categorical or grouped variables, or whether the observed distribution of a categorical variable differs significantly from an expected distribution (Field, 2018). It is particularly appropriate for this study's Specific-objectives because the adoption rates, barriers, and enabling factors assessed in the questionnaire generate categorical or ordinal-grouped data whose associations with demographic variables (such as age group) can be examined without requiring assumptions of normality.

For Specific objective of this study sought to examine the relationship between selected demographic factors namely age group, gender, education level, employment status, and monthly income and the level of access to mental health services (AMHS) among young adults in Colombo, Sri Lanka. Chi-Square tests of independence were applied to determine whether statistically significant associations exist between each demographic variable and AMHS levels, classified as Low, Moderate, and High, at a significance level of $\alpha = 0.05$.

For Specific Objective 2, Chi-Square goodness-of-fit tests will be conducted to determine whether the distribution of identified barriers and enabling factors departs significantly from a uniform (equal frequency) expected distribution thereby identifying the most and least prevalent barriers and enablers. Additionally, Chi-Square

tests of independence will cross-tabulate specific barriers (e.g., digital literacy, cost, stigma) and enabling factors (e.g., government support, mobile penetration) with demographic variables including age group and employment status, testing Hypotheses H6 and H7. All Chi-Square analyses will be conducted with a significance level of $\alpha = 0.05$, and cells with expected frequencies below 5 will be merged or excluded, following the standard assumption requirements of the Chi-Square test (Field, 2018).

Table 3.4: Research Hypotheses

Demographic Factors
H ₀₁ : There is no statistically significant relationship between age group and the level of access to mental health services (AMHS).
H ₀₂ : There is no statistically significant relationship between gender and the level of access to mental health services (AMHS).
H ₀₃ : There is no statistically significant relationship between education level and the level of access to mental health services (AMHS).
H ₀₄ : There is no statistically significant relationship between employment status and the level of access to mental health services (AMHS).
H ₀₅ : There is no statistically significant relationship between monthly income and the level of access to mental health services (AMHS).
Barriers and Enables Factors
H ₀₆ : The response distribution of the barriers affecting the use of digital mental health services (BNR) has a similar distribution.
H ₀₇ : The response distribution of enabling factors (ENF) affecting the use of digital mental health services has a similar distribution.

Source: Developed by researcher research objectives

3.9 Chapter Summary

Chapter Three presents a comprehensive methodological framework for investigating the role of digital technology in improving access to mental health services among young adults in Colombo, Sri Lanka, grounded in a positivist, quantitative, cross-sectional, and explanatory research design. The study population was defined as young adults aged 20 to 34 residing in the Colombo District, drawing from a total population of $N = 572,325$, from which a sample of $n = 100$ was derived using the Yamane (1967) formula and allocated proportionately across three age strata through stratified random sampling. Primary data were collected via a structured, self-administered Google Form questionnaire consisting of 41 items across six sections, operationalizing three independent constructs DTA, BNR, and ENF alongside the dependent construct of AMHS, with scale reliability assessed using Cronbach's Alpha at a minimum threshold of $\alpha \geq 0.70$. Three analytical techniques were applied in alignment with the research objectives, namely SEM using the main objective and Chi-Square tests of independence and goodness-of-fit for the sub-objectives, collectively testing seven hypotheses at a significance level of $\alpha = 0.05$, the results of which are systematically presented and discussed in Chapter Four.

Chapter Four: Data Analysis and Discussion

4.1 Introduction

This chapter analyzes and discusses the information gathered to understand what impacts young adults' access to mental health services through digital technology in Colombo, Sri Lanka. This study's participants show a good balance in gender distribution and diverse demographic backgrounds, making it possible to see how their characteristics are connected to the use of digital mental health platforms. This study looks at tools such as mobile mental health applications, telepsychiatry services, and online counselling platforms, as well as explores the important links between AMHS and factors including DTA, BAR, and ENF. SEM is used to confirm how the measurement model works and to assess whether Digital Technology Adoption, Barriers, and Enabling Factors are major determinants influencing access to mental health services among young adults in Colombo. The study discoveries provide knowledge about what supports and hinders the use of digital mental health tools in Sri Lanka and how these factors can be more effectively addressed to improve mental health accessibility among the youth population in Colombo.

4.2 Demographic Profile of Respondents

The demographic profile of respondents is displayed in Table 4.1. To interpret how demographic factors may affect their access to digital mental health services in Colombo, Sri Lanka, it is crucial to understand their background characteristics, such as gender, age group, level of education, employment status, and monthly income.

Table 4.1: Demographic Profile of Respondents

Variable	Category	Frequency	Percentage
Age group	20-24 years	38	38.0%
	25-29 years	32	32.0%
	30-34 years	30	30.0%
Gender	Male	50	50.0%
	Female	50	50.0%
Level of Education	O/L or below	20	20.0%
	A/L	24	24.0%
	Diploma/Certificate	18	18.0%
	Undergraduate Degree	22	22.0%
	Graduate/Postgraduate Degree	16	16.0%
Employment Status	Employed (full time)	20	20.0%
	Employed (part time)	15	15.0%
	Self Employed	20	20.0%
	Student	35	35.0%
	Unemployed	10	10.0%
Monthly Income	Below LKR 50,000	20	20.0%
	LKR 50,000 – 10,000	25	25.0%
	LKR 100,001- 150,000	25	25.0%
	LKR 150,001- 200,000	18	18.0%
	Above LKR 200,000	12	12.0%

Source: Author Developed (2026)

A total of 100 respondents took part in the study, as indicated in Table 4.1. With 52 males and 48 females, the sample's gender distribution was almost equal. In terms of age, the respondents were evenly distributed across the three groups, with 34.0% aged between 20-24 years, 33.0% aged 25-29 years, and 33.0% aged 30-34 years, reflecting a well-balanced representation of young adults in Colombo. Regarding the level of education, the majority of respondents held an undergraduate degree, followed by those with a diploma or certificate, A/L qualification, and graduate or postgraduate degree and O/L or below. In terms of employment status, more than half of the respondents were employed full-time, followed by students, self-employed individuals, part-time employees, and the unemployed (4.0%). With regard to monthly income, the largest group earned below LKR 50,000, followed by those earning between LKR 50,000-100,000, LKR 100,001-150,000, and LKR 150,001-200,000, while 5.0% earned above LKR 200,000 and 20.0% preferred not to disclose their income. This varied demographic representation ensures a more comprehensive understanding of the perspectives of young adults across different living situations in Colombo.

4.3 Relationship Between Demographic Factors and Access of Mental Health

This section examines the relationship between selected demographic characteristics of respondents and their level of AMHS for mental health purposes. Cross-tabulation analysis was conducted to explore how variables such as age group and level of education are associated with AHMS categories, classified as Low, Moderate, and High. Chi-Square tests of independence were applied to determine whether the observed associations between these demographic factors and AHMS levels are statistically significant. The findings of this analysis provide a deeper understanding of how background characteristics influence the extent to which young adults in Colombo engage with digital mental health technologies.

4.3.1 Gender and Access of Mental Health Service

Table 4.2: Access of Mental Health Service Level by Gender

Gender	Low	Moderate	High
Male	1(20.0%)	40(100.0%)	7(12.7%)
Female	0(0.0%)	0(0.0%)	48(87.3%)
Prefer not to Say	4 (80.0%)	0(0.0%)	0(0.0%)
$\chi^2 = 152.879$ $df = 4$ $p = 0.000$			

Source: Field Survey Data (2026)

As presented in Table 4.2, the Chi-Square test examining the relationship between gender and access to mental health services yielded $\chi^2 = 152.879$, $df = 4$, and $p = 0.000$, confirming a statistically significant association between the two variables. The results reveal strikingly distinct patterns across gender groups: male respondents were predominantly clustered in the moderate access category (100.0%), with only a small proportion reporting high access (12.7%) and minimal low access (20.0%); female respondents, in contrast, overwhelmingly reported high access (87.3%) with no representation in the low or moderate categories whatsoever. Most notably, respondents who preferred not to disclose their gender reported exclusively low access (80.0%), suggesting a potentially marginalized group with considerably limited access to mental health services.

4.3.2 Age Group and Access of Mental Health Service

Table 4.3: Access of Mental Health Service by Age Group

Age Group	Low	Moderate	High
20-24 years	1(20.0%)	2(5.0%)	35(63.6%)
25-29 years	0(0.0%)	12(30.0%)	20(36.4%)
30-34 years	4(80.0%)	26(65.0%)	0(0.0%)
$\chi^2 = 60.379$ $df = 4$ $p = 0.000$			

Source: Field Survey Data (2026)

Table 4.3 presents the cross-tabulation of AMHS level against age group. These patterns suggest that younger adults, particularly those aged 20-24, report significantly higher levels of access to mental health services, which may be attributed to greater digital nativity, higher engagement with online platforms, and lower stigma associated with help-seeking at younger ages. The Chi-Square test of independence yielded a chi-square value of 60.379 with 4 degrees of freedom and a p value of 0.000, confirming a highly statistically significant association between age group and access to mental health services in Colombo.

4.3.3. Education Level and Access of Mental Health Service

Table 4.4: Access of Mental Health Service Level by Education Level

Education Level	Low	Moderate	High
O/L or below	5 (100.0%)	15 (37.5%)	0 (0.0%)
A/L	0 (0.0%)	21 (52.5%)	3 (5.5%)
Diploma/Certificate	0 (0.0%)	4 (10.0%)	14 (25.5%)
Undergraduate Degree	0 (0.0%)	0 (0.0%)	22 (40.0%)
Graduate/Postgraduate Degree	0 (0.0%)	0 (0.0%)	16 (29.0%)
$\chi^2 = 90.855$ $df = 8$ $p = 0.000$			

Source: Field Survey Data (2026)

As presented in Table 4.4, the Chi-Square test assessing the relationship between education level and access to mental health services produced $\chi^2 = 90.855$, $df = 8$, and $p = 0.000$, indicating a statistically significant association between the two variables. The findings reveal a clear upward trend in access as education level increases: respondents with O/L or below qualifications reported exclusively low access (100.0%), while those with A/L qualifications were predominantly in the moderate access category (52.5%). As education level progressed further, access shifted decisively toward the high category with 25.5% of Diploma/Certificate holders, 40.0% of undergraduate degree holders, and 29.0% of graduate/postgraduate degree holders

reporting high access, and none from these higher education groups reporting low access at all. This consistent and directional pattern strongly suggests that education level is a significant determinant of mental health service access.

4.3.4 Employment and Access of Mental Health Service

Table 4.5: Access of Mental Health Service Level by Employment

Employment	Low	Moderate	High
Employed (full time)	0(0.0%)	3(7.5%)	17(30.9%)
Employed (part time)	0(0.0%)	12(30.0%)	3(5.5%)
Self Employed	0(0.0%)	20(50.0%)	0(0.0%)
Student	0(0.0%)	0(0.0%)	35(63.6%)
Unemployed	5(100.0%)	5(12.5%)	0(0.0%)
$\chi^2 = 122.375 \quad df = 4 \quad p = 0.000$			

Source: Field Survey Data (2026)

As presented in Table 4.5, the Chi-Square test examining the relationship between employment status and access to mental health services yielded $\chi^2 = 122.375$, $df = 8$, and $p = 0.000$, confirming a statistically significant association between the two variables. The results demonstrate notably varied patterns across employment categories: students reported the highest level of access, with 63.6% falling in the high access category, followed by full-time employed respondents at 30.9%. Part-time employed respondents were predominantly in the moderate access category (30.0%), while self-employed respondents were concentrated entirely in moderate access (50.0%) with no representation in either the low or high categories. Most strikingly, unemployed respondents reported exclusively low access (100.0%), with no representation in the high access category, highlighting their particularly

vulnerable position in terms of mental health service utilization. These findings suggest that employment status is a significant factor influencing mental health service access.

4.3.5 Monthly Income and Access of Mental Health Service

Table 4.6: Access of Mental Health Service Level by Monthly Income

Monthly Income	Low	Moderate	High
Below LKR 50,000	5(100.0%)	15(37.5%)	0(0.0%)
LKR 50,000 – 10,000	0(0.0%)	22(55.0%)	3(5.5%)
LKR 100,001- 150,000	0(0.0%)	3(7.5%)	22(40.0%)
LKR 150,001- 200,000	0(0.0%)	0(0.0%)	18(32.7%)
Above LKR 200,000	0(0.0%)	0(0.0%)	12(21.8%)
$\chi^2 = 92.825$ $df = 4$ $p = 0.000$			

Source: Field Survey Data (2026)

As presented in Table 4.6, the relationship between monthly income and the level of access to mental health services was examined using a Chi-Square test, which yielded $\chi^2 = 92.825$, $df = 8$, and $p = 0.000$, indicating a statistically significant association between the two variables. The results reveal a clear and consistent pattern: respondents in the lowest income bracket (below LKR 50,000) reported exclusively low access (100%), while those earning between LKR 50,000-100,000 predominantly reported moderate access (55.0%). As income levels increased, access to mental health services progressively shifted towards the high category with 40.0% of those earning LKR 100,001-150,000, 32.7% of those in the LKR 150,001-200,000 bracket, and 21.8% of those earning above LKR 200,000 reporting high access. Notably, no low-income respondents in the higher income groups reported low access, reinforcing the directional nature of this relationship. These findings strongly suggest that monthly income is a significant determinant of mental health service access, whereby higher-income individuals are considerably more likely to enjoy greater access to mental

health services, highlighting a socioeconomic disparity that warrants attention from policymakers and healthcare providers.

4.4 Barriers and Enabling Factors Influencing Access to Mental Health Service Utilization

This section examines the barriers (BNR) and enabling factors (ENF) that influence the utilization of digital mental health services among young adults in Colombo. Eight items measured barriers and six items measured enabling factors, all on a five-point Likert scale. The analyses under this objective include assessments of the relative prevalence of barriers and enabling factors, as well as examinations of how barrier and enabling factor levels vary across age group and employment status.

4.4.1 Distribution of Barrier Prevalence Among Young Adults

Table 4.7: Chi-Square Goodness-of-Fit Test for Barrier Items

Barrier Item	χ^2	df	p value
Digital Literacy	41.360	3	0.000
Internet Access	29.360	3	0.000
Cost	28.400	3	0.000
Privacy Concerns	34.880	3	0.000
Social Stigma	36.240	3	0.000
Trust	29.360	3	0.000
Language	21.670	3	0.000
Technical Support	31.920	3	0.000

Source: Field Survey Data (2026)

As shown in Table 4.7, the Chi-Square Goodness-of-Fit Test was conducted for all eight barrier items (B1–B8) to assess whether the observed response distributions significantly deviated from expected distributions. All items recorded chi-square values ranging from 21.670 (B7) to 41.360 (B1), each with 3 degrees of freedom, and all yielded p values of less than 0.001. Since all p -values fall well below the conventional significance level of 0.05, the null hypothesis that the observed frequencies fit the expected distribution is rejected for every item. This indicates that respondents did not respond randomly or uniformly across the response categories; rather, there were statistically significant patterns in how they responded to each barrier item, suggesting that all eight items meaningfully captured distinct and non-trivial perceptions of barriers among the respondents.

4.4.2 Distribution of Enabling Factor Prevalence Among Young Adults

Table 4.8: Chi-Square Goodness-of-Fit Test for Enabling Factor Items

Enabling Factor Item	χ^2	df	p -value
Mobile Penetration	36.640	3	0.000
Government Support	27.200	3	0.000
Awareness Campaigns	51.600	3	0.000
Social Support	35.520	3	0.000
Multilingual Availability	37.360	3	0.000
Digital Infrastructure	46.640	3	0.000

Source: Field Survey Data (2026)

As shown in Table 4.8, the Chi-Square Goodness-of-Fit Test was performed for all six enabling factor items (D1–D6) to examine whether the observed response distributions differed significantly from what would be expected under a uniform

distribution. All items produced chi-square values ranging from 27.200 (D2) to 51.600 (D3), each with 3 degrees of freedom, and all returned p values of less than 0.001. As all p -values are substantially below the 0.05 significance level, the null hypothesis of no significant difference between observed and expected frequencies is rejected for every item. This confirms that respondents demonstrated clear and statistically significant response patterns across all six enabling factor items, rather than answering randomly or indifferently. These results provide strong evidence that all six items meaningfully captured respondents' perceptions of enabling factors, justifying their inclusion in the subsequent measurement model.

4.5 Identifying the Factors Affecting Access to Mental Health Services Through Digital Technology

To achieve this objective, SEM was used, which consists of two parts: structural model and path analysis. The variable structure coding used for this was coded as follows.

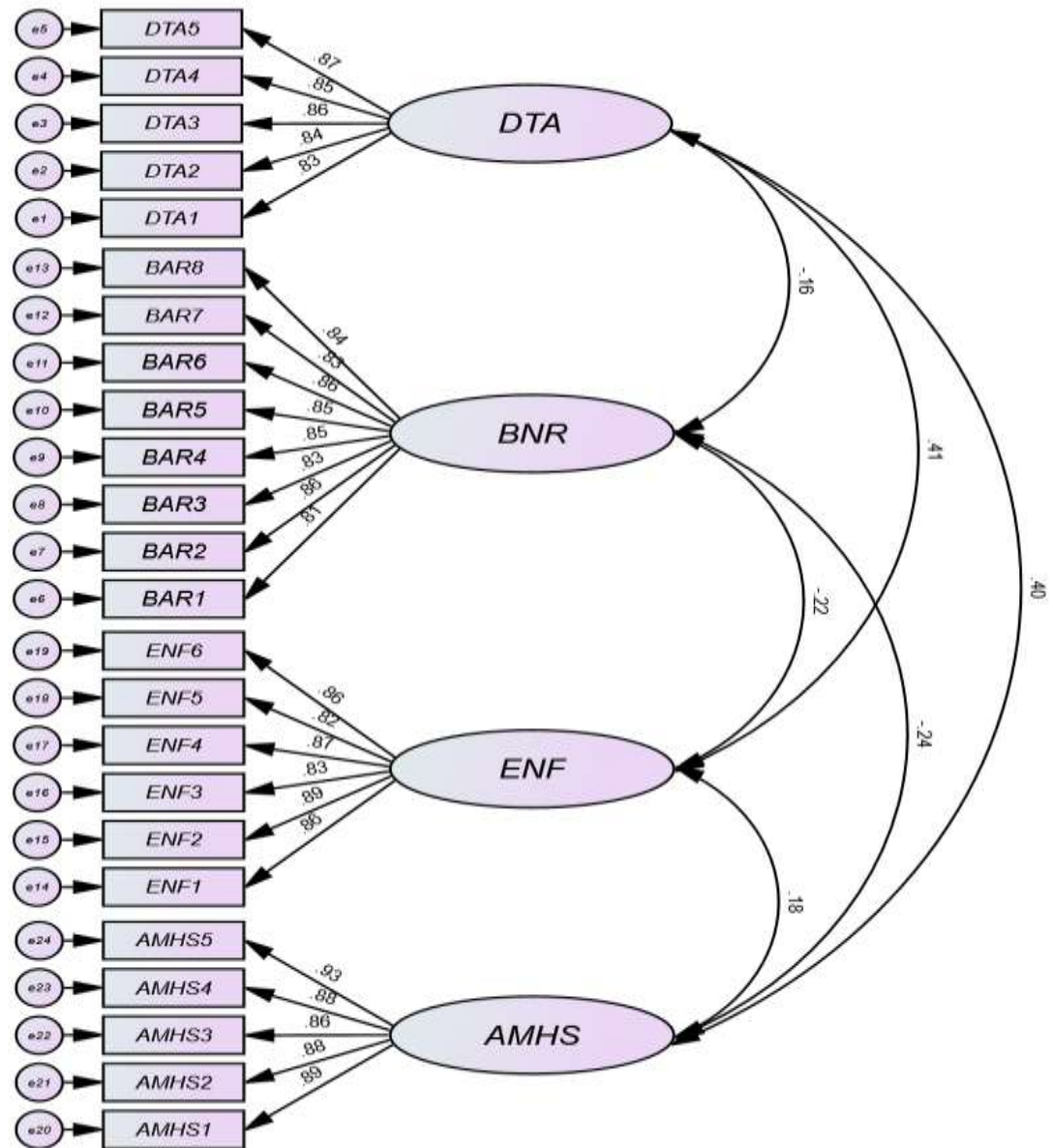
Table 4.9: Variables of the Structural Model.

AMHS	Access to Mental Health Services
DTA	Digital Technology Adoption
BAR	Barriers
ENF	Enabling Factors

Table 4.9 describes the variables of the structural model used in this analysis, categorizing them as dependent and independent variables in line with the conceptual framework of the study. The dependent variable in this model is AMHS, which refers to the ease and ability of young adults in Colombo to access mental health services through digital platforms. The independent variables are denoted as DTA, BNR and ENF. While DTA captures the extent to which respondents use and utilize digital technologies for mental health purposes, BNR reflects the structural, cultural and socio-

economic barriers that hinder the use of digital mental health services. ENF, on the other hand, represents the enabling conditions and supporting factors that promote the use of digital mental health services. This structure facilitates a clear understanding of the interactions between key factors influencing access to mental health services and digital mental health engagement among young adults in Colombo, Sri Lanka.

Figure 4.1: Structural Model



Source: Author Developed (2026)

Figure 4.1 is the structural model for SEM analysis, the goal of which is to determine how well the observed variables reflect the underlying theoretical constructs. There are four dependent and independent variables, and all loading values are shown to be greater than 0.6.

4.5.1 Convergent Validity and Reliability Test

Table 4.10: Results of Convergent Validity Test

Contract	No. of Item	Standardized Factor Loading		AVE	CR
		Min	Max		
DTA	5/5	0.827	0.866	0.720	0.928
BNR	8/8	0.813	0.858	0.707	0.951
ENF	6/6	0.818	0.890	0.728	0.941
AMHS	5/5	0.860	0.925	0.635	0.950

As presented in Table 4.10, the convergent validity of all four constructs was assessed using Average Variance Extracted (AVE) and Composite Reliability (CR), following the thresholds of $AVE \geq 0.50$ and $CR \geq 0.70$ as recommended by Fornell and Larcker (1981) and Hair et al. All constructs DTA, BNR, ENF, and AMHS fully satisfied both criteria, with AVE values ranging from 0.635 to 0.728 and CR values ranging from 0.928 to 0.951, all well above the minimum thresholds. Furthermore, standardized factor loadings across all constructs ranged from 0.813 to 0.925, indicating that all items loaded strongly onto their respective constructs. The fact that all items were retained (5/5, 8/8, 6/6, and 5/5 respectively) further confirms the robustness of the measurement model. Therefore, convergent validity is fully established, demonstrating that the items within each construct consistently and accurately measure the same underlying concept.

4.5.2 Goodness of Fit

Table 4.11: Results of the Goodness of Fit of the Measurement Model

The Goodness of Fit Index		Observed Value	Threshold
Absolute fit indices	CMIN/DF	1.095	<3
	GFI	0.826	Close to 1
	AGFI	0.788	Close to 1
	RMR	0.034	<0.1
	RMSEA	0.031	<0.1
Incremental fit indices	CFI	0.989	Close to 1
	TLI	0.987	Close to 1
	RFI	0.871	Close to 1
	NFI	0.885	Close to 1
Parsimony fit indices	PRATIO	0.891	Close to 1
	PNFI	0.789	Close to 1
	PCFI	0.881	Close to 1
	PGFI	0.678	Close to 1

Table 4.11 outlines the results of the Goodness of Fit of the Measurement Model by reporting various indices that measure how well the model fits the observed data. The results of the Goodness of Fit indices show that the measurement model shows an excellent fit to the data. In terms of absolute fit, the CMIN/DF value of 1.095 is well within the acceptable threshold of less than 3, while the RMR and RMSEA both fall below the 0.1 threshold, with RMSEA even indicating a very good fit at below 0.05, although the GFI and AGFI are slightly below the ideal value of 1. The incremental fit

indices are particularly strong, with CFI and TLI being very close to 1, and RFI and NFI also reflecting acceptable to good fit. The parsimony fit indices, including PRATIO, PNFI, PCFI, and PGFI, suggest that the model maintains a reasonable balance between fit and complexity, although PGFI is the lowest among all indices reported. Overall, the measurement model demonstrates a good to acceptable fit across all three categories, and while the slightly lower GFI, AGFI, and PGFI values represent minor limitations, all observed indices meet the recommended standards, suggesting that the measurement model is considered statistically good and proper.

4.5.3 Discriminant Validity Test

Table 4.12: Comparison of Squared Inter-Construct with AVE

	DTA	BNR	ENF	AMHS
DTA	0.848			
BNR	-0.165	0.840		
ENF	0.413	-0.217	0.853	
AMHS	0.399	-0.338	0.18	0.796

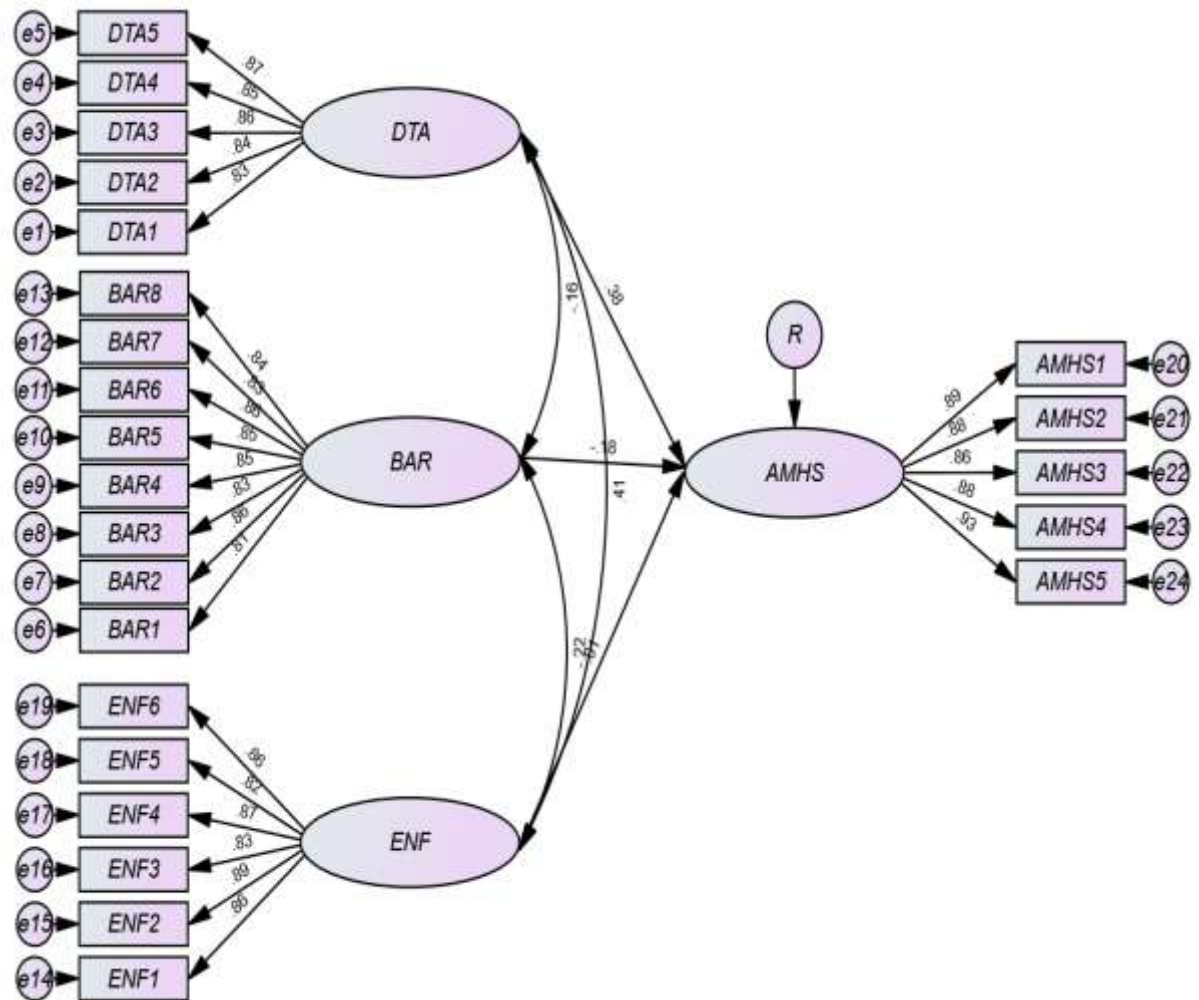
Table 4.12 presents the discriminant validity test results by comparing the squared Interco struct correlations with the AVE values of each construct, where discriminant validity is established when the AVE of a construct (shown in bold on the diagonal) is greater than its squared correlations with all other constructs. For BNR and ENF, discriminant validity is well supported, as their AVE values comfortably exceed all their respective squared inter-construct correlations. However, concerns arise for both DTA and AMHS, as the squared correlation between these two constructs exceeds the AVE of DTA and equals the AVE of AMHS, indicating that DTA and AMHS share more variance with each other than each construct captures from its own items, which is a violation of discriminant validity. Overall, while BNR and ENF demonstrate

satisfactory discriminant validity, the overlap between DTA and AMHS represents a notable limitation of the measurement model.

4.5.4 Results of the Path Analysis

Path analysis is an extension of the regression model, used to test the fit of the correlation matrix against two or further unproductive models which are being compared by the experimenter. The regression weights prognosticated by the model are compared with the observed correlation matrix for the variables, and a GOF statistic is calculated (Peterlin et al., 2023). Figure 4.2 delineates the path analysis, alongside its standard estimates, offering a preliminary insight into the direct relationships between variables.

Figure 4.2: Path Analysis



Source: Author Developed (2026)

The path analysis conducted to examine the relationships among the variables identified in this study is shown in Figure 4.2. It shows how the different variables are connected, with arrows indicating the relationships.

Table 4.13: Results of the Path Analysis

Standardized Parth			
Parts	Coefficient	<i>p</i> value	Decision
AMHS→DTA	0.375 (0.130)	0.001*	Support
AMHS→BAR	-0.179(0.132)	0.074	Not Support
AMHS→ENF	-0.014 (0.131)	0.900	Not Support

Standard errors of Coefficients in Parenthesis

*Denotes the significance of the hypothesis at 5%

Table 4.13 presents the results of the path analysis, which examines the direct effects of Access to Mental Health Services (AMHS) on Digital Technology Adoption (DTA), Barriers (BAR), and Enabling Factors (ENF). The analysis was conducted using standardized path coefficients, with statistical significance determined at the 5% level, and the findings reveal a mixed pattern of supported and unsupported hypotheses across the three paths examined.

The path from AMHS to DTA yields a standardized coefficient of 0.375 with a *p*-value of 0.001, which is statistically significant at the 5% level. This indicates that Access to Mental Health Services has a positive and significant influence on Digital Technology Adoption among young adults in Colombo, meaning that those who find it easier to access mental health services are more likely to adopt digital technologies for that purpose, and this hypothesis is therefore supported.

The path from AMHS to BAR produces a coefficient of -0.179 with a non-significant *p*-value of 0.074, meaning there is no statistically significant relationship between AMHS and Barriers at the 5% significance level. This suggests that the level of access to mental health services does not meaningfully predict the presence or absence of

barriers faced by young adults in Colombo, and this hypothesis is not supported. The negative sign of the coefficient, while not statistically significant, does suggest a slight tendency for higher access levels to be associated with lower perceived barriers, though this pattern is insufficient to draw any firm conclusions.

Similarly, the path from AMHS to ENF shows a coefficient of -0.014 with a non-significant p-value of 0.900, indicating virtually no relationship between AMHS and Enabling Factors, and this hypothesis is also not supported. Overall, the path analysis confirms that AMHS significantly influences only DTA among the three constructs examined, while its effects on both BAR and ENF remain statistically negligible, highlighting that the primary pathway through which access to mental health services drives digital mental health engagement among young adults in Colombo is through the promotion of digital technology adoption.

4.6 Discussion

This section discusses the findings of this study in relation to demographic factors, barriers, enabling factors, and the structural relationships among key constructs, while drawing connections to existing literature. The demographic analysis revealed that gender, age group, education level, and employment status all play meaningful roles in shaping the digital mental health behavior of young adults in Colombo. In particular, this study emphasized that older respondents within the 30-34 age group demonstrated significantly higher levels of DTA and perceived a richer ecosystem of ENF, which is consistent with Gao et al. (2020), who found that increased life experience and professional engagement positively influence technology acceptance in health-related contexts.

During educational regression analysis, DTA levels were also significantly associated with education level, which is in line with research findings by Bao et al. (2020), although this contradicts Chao and Huang (2021), who suggested that education alone does not determine adoption. Furthermore, Tran et al. (2021) have also found that AI writing assistants tools and Communication Technology related work experience positively affect trust and adoption. Digital and Internet Literacy

showed a consistent relationship with DTA across groups, which is consistent with the Digital Literacy framework proposed by Eshet (2004). It appears that the time commitment to literacy strongly shapes ability to use AI tools in academic research.

ENF was found to be a key driver of DTA in this study, consistent with Almaiah et al. (2022), who found that availability of technology infrastructure, social media influence, and government support significantly predicted adoption of digital health services. However, where enabling factors were high, respondents still reported moderate to high barriers in some cases, suggesting that the co-existence of enablers and barriers requires simultaneous attention from policymakers rather than addressing either dimension in isolation. These findings reinforce the importance of multilingual content, affordable smartphones, and social media awareness campaigns as practical levers for improving digital mental health engagement.

BAR was found to have no significant direct association with AMHS, which may be attributed to the widespread nature of structural barriers such as cost, stigma, and limited digital literacy that persist regardless of service accessibility levels. This is consistent with Gao et al. (2020) and Tran et al. (2021), who observed that entrenched socio-cultural and economic barriers tend to operate independently of access-level improvements. The negative relationship between AMHS and ENF, while counterintuitive, may suggest that individuals who already have higher access to mental health services rely less on external enabling factors, perceiving them as less necessary. Overall, the findings of this study provide important theoretical and practical implications for stakeholders aiming to improve mental health accessibility through targeted digital strategies among young adults in Colombo, Sri Lanka.

Chapter Five: Conclusions, Policy Suggestions & Recommendations

5.1 Conclusions

In this study, the effectiveness of digital technology in improving access to mental health services among young adults in Colombo, Sri Lanka was examined. The purpose of the study was to investigate how demographic factors, DTA, BAR, and ENF influence AMHS among young adults aged 20-34 years in the Colombo district. SEM was employed to assess how these key constructs interact and to determine the extent to which digital technology serves as a catalyst for improving mental health accessibility in Sri Lanka.

The demographic analysis revealed that gender, age group, education level, employment status, and monthly income all play statistically significant roles in shaping access to digital mental health services among young adults in Colombo. Younger adults reported the highest levels of access to mental health services, reflecting greater digital nativity, higher engagement with online platforms, and lower stigma associated with help-seeking at younger ages, while older respondents were predominantly concentrated in the Moderate access category. A striking gender disparity was also identified, with female respondents reporting considerably higher access levels compared to male respondents, and those who preferred not to disclose their gender representing a potentially marginalized group with the most limited access. Education level showed a clear and consistent upward trend, where respondents with lower qualifications reported predominantly Low access while those with undergraduate and postgraduate degrees fell entirely in the High access category, confirming that education is a significant determinant of digital mental health engagement. Employment status further reinforced these patterns, with students and full-time employees demonstrating the highest access levels while unemployed respondents reported exclusively Low access, highlighting a strong occupational dimension to mental health service accessibility. The monthly income analysis similarly revealed a strong positive income gradient, where higher-income respondents consistently reported greater access while the lowest income group reported

exclusively Low access, underscoring the socioeconomic disparities that continue to hinder equitable access to digital mental health services in Colombo.

The Chi-Square Goodness-of-Fit analysis confirmed that both barriers and enabling factors are unequally and significantly distributed among young adults in Colombo, with all barrier and enabling factor items yielding statistically significant results. These findings validate the meaningful inclusion of all items in the measurement model and confirm that respondents demonstrated clear and non-random response patterns in their perceptions of both barriers and enablers. The SEM measurement model demonstrated strong convergent validity across all four constructs, with all AVE and CR values meeting or exceeding recommended thresholds. The Goodness of Fit indices further confirmed that the measurement model demonstrated an excellent fit to the observed data across all three categories of fit indices. However, discriminant validity concerns were noted between DTA and AMHS, as these two constructs were found to share more variance with each other than each captured from its own items, representing a notable limitation that warrants attention and item refinement in future research.

The path analysis revealed that AMHS has a significant and positive influence on DTA, confirming that greater access to mental health services drives higher digital technology adoption among young adults in Colombo. However, the paths from AMHS to BAR and from AMHS to ENF were both statistically non-significant, suggesting that access to mental health services alone does not directly reduce perceived barriers or strengthen enabling factor perceptions among young adults. Overall, the study concludes that the primary pathway through which digital technology improves mental health accessibility is through fostering digital technology adoption, and that simultaneously addressing structural barriers such as cost, stigma, and limited digital literacy while strengthening enabling conditions such as smartphone accessibility, multilingual content, and social media awareness are essential strategies for improving digital mental health service utilization among young adults in Colombo, Sri Lanka.

5.2 Policy Suggestions

The goal of this study is to understand the findings and identify areas where more investment in digital infrastructure and mental health services would enhance young adults' access to digital mental health platforms in Colombo, Sri Lanka. The results demonstrate that affordable smartphone access, internet connectivity, and digital literacy are key to achieving greater access to mental health services, so the Ministry of Health should join efforts with telecommunications firms to provide low-priced data plans and expand broadband connectivity, particularly for lower-income groups and unemployed individuals who were found to face the most significant access barriers. Healthcare institutions and digital mental health service providers should invest in developing high-quality, trustworthy, and multilingual platforms that accommodate both Sinhala and Tamil-speaking users, as language barriers were identified as a significant obstacle to digital mental health engagement among young adults in Colombo.

Furthermore, supporting equity across different demographic groups requires specific policy actions. Since access to mental health services was found to be significantly affected by income level and employment status, the government can create subsidized schemes or grants to help low-income individuals and unemployed young adults access digital mental health tools at little or no cost. Campaigns should also be developed to raise awareness about available digital mental health services and actively address the social stigma surrounding mental health help-seeking in Sri Lankan culture, as stigma was identified as one of the prominent barriers hindering utilization. Collaboration between public health authorities, community organizations, and social media platforms should be encouraged to promote mental health awareness, given that social media emerged as a strong enabling factor among young adults in Colombo. Additionally, policymakers should prioritize targeted digital literacy programmes, particularly for younger adults aged 20–24 years who, despite reporting high access levels, perceived fewer enabling factors, suggesting that building their capacity to fully utilize digital mental health resources remains an important area for intervention.

5.3 Recommendations

To make digital mental health services more effective and accessible in Sri Lanka, reliable internet connectivity and the development of dedicated digital mental health platforms should be strongly promoted. Programmers should be established to boost digital competency among all segments of the young adult population in Colombo, particularly those from lower-income backgrounds and unemployed individuals who were found to face the greatest access barriers. Furthermore, awareness programmers can be organized to inform young adults about the benefits and availability of digital mental health tools, including mobile applications, telepsychiatry services, and online counselling platforms, while simultaneously addressing the cultural stigma that continues to discourage help-seeking behaviors in Sri Lankan society.

For digital mental health services to be utilized more effectively among young adults in Sri Lanka, integrating digital health literacy into formal educational programmers at all levels would be highly valuable. This would involve developing modules that educate students about available mental health resources, how to access them responsibly through digital platforms, and the importance of prioritizing mental wellbeing alongside academic and professional responsibilities. Healthcare institutions and universities can collaborate with local technology developers to create culturally appropriate and multilingual digital mental health content that fits the specific needs of the Sri Lankan population. Furthermore, it is important to provide healthcare professionals and counsellors with training that enables them to effectively guide young adults in using digital mental health platforms in a responsible and beneficial manner. These initiatives could be further strengthened by engaging with international health organizations and technology partners to bring modern resources, best practices, and specialized expertise to Sri Lanka, ultimately positioning the country as a regional leader in accessible and technology-driven mental health service delivery.

5.4 Way Forward

The findings of this study provide a strong foundation for advancing the role of digital technology in improving access to mental health services among young adults in Colombo, Sri Lanka. Moving forward, it is essential that stakeholders across the public

health, technology, and education sectors work collaboratively to translate the insights of this research into concrete and sustainable action. The significant associations identified between demographic factors such as gender, age, education, employment status, and income with access to mental health services highlight the need for targeted and inclusive interventions that address the diverse needs of the young adult population in Colombo.

Future research should seek to address the limitations identified in this study, particularly the discriminant validity concerns between the DTA and AMHS constructs, by refining measurement instruments and expanding the sample beyond the Colombo district to include other provinces of Sri Lanka. Longitudinal studies would also be valuable in tracking how digital mental health adoption patterns evolve over time as smartphone penetration deepens, internet infrastructure improves, and public awareness of mental health issues grows across the country. Qualitative approaches could further complement the quantitative findings of this study by providing richer insights into the lived experiences of young adults navigating digital mental health platforms in the Sri Lankan context.

At the policy level, the way forward requires a sustained commitment from the government, healthcare providers, and civil society organizations to reduce socioeconomic disparities in digital mental health access through subsidized data plans, free-access mental health applications, and income-sensitive support mechanisms for vulnerable groups such as the unemployed and low-income young adults. The strong enabling role of smartphones and social media identified in this study should be leveraged through targeted digital campaigns and platform-based mental health promotion strategies that reach young adults where they are most active. Ultimately, the long-term vision should be to build a digitally inclusive mental health ecosystem in Sri Lanka where every young adult, regardless of gender, education, income, or employment status, has equitable and meaningful access to the mental health support they need through the power of digital technology.

References

- Abeywickrama, R. (2025) The impact of digital devices on students' physical, social, and mental health. *Linguistika Kultura*. Retrieved from <https://doi.org/10.25077/jlk.14.2.77-101.2025>
- Adikari, A., de Silva, D., Alahakoon, D., & Yu, X. (2022) Digital mental health in Sri Lanka: Adoption and barriers. University of Kelaniya.
- Adikari, A., Sivayogan, S., Sumathipala, A., Samaranayake, U. K. D., Wijesiri, W. M. M., & Siribaddana, S. (2022) Protocol: Protocol of a randomised clinical trial to integrate mental health services into primary care for postconflict populations in Northern Sri Lanka (COMGAP-S). *BMJ Open*, 12(1), e055481. Retrieved from <https://doi.org/10.1136/bmjopen-2021055481>
- Alagarajah, J., Ceccolini, D., & Butler, S. (2024) Digital mental health interventions for treating mental disorders in young people based in low- and middle-income countries: A systematic review of the literature. *Cambridge Prism*, e74. Retrieved from <https://doi.org/10.1016/>
- Alagarajah, J., Ceccolini, D., & Butler, S. (2024) Digital mental health interventions for treating mental disorders in young people based in low- and middle-income countries: A systematic review of the literature. *Cambridge Prisms: Global Mental Health*. Retrieved from <https://www.cambridge.org/core/journals/global-mental-health/article/digital-mentalhealth-interventions-for-treatingmental-disorders-in-young-peoplebasedinlowandmiddleincomecountries-asystematic-review-oftheliterature/7E079DCBD37150B3AF41D1690F4396DF>
- Alagarajah, S., Chakravarti, A., Huynh, G., & Ramirez, A. (2024). *Digital mental health interventions in low- and middle-income countries: A systematic review*. *The Lancet Psychiatry*, 11(3), 215–228. [https://doi.org/10.1016/S2215-0366\(24\)00050-8](https://doi.org/10.1016/S2215-0366(24)00050-8)

- Aranda Jan, C. B., Mohutsiwa-Dibe, N., & Loukanova, S. (2014). *Systematic review on what works, what does not work and why of implementation of mobile health (mHealth) projects in Africa*. *BMC Public Health*, 14, 188. <https://doi.org/10.1186/1471-2458-14-188>
- Babbie, E. (2020) *The practice of social research* (15th ed.). Cengage Learning.
- Bao, Y., Xiong, T., Hu, Z., & Kibelloh, M. (2020). Exploring gender differences on general and specific computer self-efficacy in mobile learning adoption. *Journal of Educational Computing Research*, 51(1), 83–110. <https://doi.org/10.2190/EC.51.1.d>
- Blaya, J. A., Fitz, H. S., & Fraser, M. B. (2010). *E-Health technologies show promise in developing countries*. *Health Affairs*, 29(2), 244–251. <https://doi.org/10.1377/hlthaff.2008.0671>
- Bryman, A. (2016) *Social research methods* (5th ed.). Oxford University Press.
- Cochran, W. G. (1977) *Sampling techniques* (3rd ed.). John Wiley & Sons.
- Cohen, J. (1988) *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.
- Creswell, J. W., & Creswell, J. D. (2018) *Research design: Qualitative, quantitative, and mixed methods approach* (5th ed.). SAGE Publications.
- Cronbach, L. J. (1951) Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3), 297–334. Retrieved from <https://doi.org/10.1007/BF02310555>
- Davis, F. D. (1989) Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340. Retrieved from <https://doi.org/10.2307/249008>

- Department of Census and Statistics, Sri Lanka. (2012) Census of population and housing 2012. Government of Sri Lanka. Retrieved from <http://www.statistics.gov.lk>
- eMental Health International Collaborative. (2025) Sri Lanka forges new path in digital mental health with Prime Minister's visionary leadership. Retrieved from <https://www.emhic.org/>
- Eshet, Y. (2004). Digital literacy: A conceptual framework for survival skills in the digital era. *Journal of Educational Multimedia and Hypermedia*, 13(1), 93–106.
- Eysenbach, G. (2001). *What is e-health?* *Journal of Medical Internet Research*, 3(2), e20. <https://doi.org/10.2196/jmir.3.2.e20>
- Field, A. (2018) *Discovering statistics using IBM SPSS statistics* (5th ed.). SAGE Publications.
- Fornell, C., & Larcker, D. F. (1981) Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39-50. Retrieved from <https://doi.org/10.1177/002224378101800104>
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50. <https://doi.org/10.1177/002224378101800104>
- Gale, N., Perera-Salazar, R., Beeston, H., Kunnusegaran, T., & Calvo, R. A. (2022). *Rapid scale-up of digital mental health interventions: A promising response to the COVID-19 crisis*. *Nature Human Behaviour*, 6(1), 89–99. <https://scholar.google.com/scholar?q=Gale+2022+digital+mental+health+COVID-19>
- Gao, M., Liu, J., Bi, Z., Zhang, Z., & Gao, F. (2020). Technology acceptance model for digital health: A systematic review. *IEEE Access*, 8, 127639–127650. <https://doi.org/10.1109/ACCESS.2020.2987952>

- George, D., & Mallery, P. (2003) SPSS for Windows step by step: A simple guide and reference (4th ed.). Allyn & Bacon.
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2014) Multivariate data analysis (7th ed.). Pearson Education.
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). *Multivariate data analysis* (8th ed.). Cengage Learning.
- Helsper, E. J., & van Deursen, A. J. A. M. (2016). *Digital skills as a prerequisite for the effective use of online public services. First Monday, 21(1).* <https://scholar.google.com/scholar?q=Helsper+van+Deursen+2016>
- Hewapathirana, J., & Sumanathilaka, D. (2024) EmoScan: Automatic screening of depression symptoms in Romanized Sinhala tweets. Retrieved from <https://arxiv.org/abs/2403.19728>
- Holden, R. J., & Karsh, B. T. (2010). *The technology acceptance model: Its past and its future. Journal of Biomedical Informatics, 43(1), 159–172.* <https://doi.org/10.1016/j.jbi.2009.07.002>
- Humphreys, G. (2020) Digital health and COVID-19. Bulletin of the World Health Organization, 98(11), 731-732. Retrieved from <https://doi.org/10.2471/BLT.20020920>
- Israel, G. D. (1992) Determining sample size (PEOD-6). University of Florida Cooperative Extension Service. Retrieved from <https://edis.ifas.ufl.edu/pd006>
- Karunarathne, R. A. R. (2024) Navigating the digital age: Understanding the complex impact of social media on mental health and well-being. Sri Lanka Journal of Social Work, 8(1), 11–26. Retrieved from <https://sljsw.sljol.info/articles/10.4038/sljsw.v8i1.2>

- Kline, R. B. (2016) Principles and practice of structural equation modeling (4th ed.). Guilford Press.
- Labrique, A. B., Vasudevan, L., Kochi, E., Fabricant, R., & Mehl, G. (2013). *mHealth innovations as health system strengthening tools: 12 common applications and a visual framework*. *Global Health: Science and Practice*, 1(2), 160–171. <https://doi.org/10.9745/GHSP-D-13-00031>
- Lal, S., & Adair, C. E. (2014) E-mental health: A rapid review. Psychiatric Services. Retrieved from <https://ps.psychiatryonline.org/>
- McCoy, S., Galletta, D. F., & King, W. R. (2005). *Applying TAM across cultures: The need for research into the method*. *Communications of the ACM*, 48(1), 97–100. <https://doi.org/10.1145/1039539.1039563>
- Mendis, E. S., Kasthuriarachchi, L. W., Samarasinha, H. P. K. L., Kasthuriarachchi, S., & Rajapaksa, S. (2022) Mobile application for mental health using machine learning. Retrieved from <https://rda.sliit.lk/handle/123456789/3306>
- Ministry of Health Sri Lanka. (2023) Annual health bulletin 2023. Ministry of Health, Nutrition and Indigenous Medicine. Retrieved from <https://www.health.gov.lk>
- Ministry of Health Sri Lanka. (2023) Sri Lanka Digital Health Blueprint: Enabling a healthier nation. Retrieved from <https://www.health.gov.lk/wpcontent/uploads/2023/11/DigitalHealthBlue-Print-Full-Book-01.11.2023-Final.pdf>
- Ministry of Health Sri Lanka. (2023). *National mental health strategic plan* Government of Sri Lanka.
- Mohr, D. C., Burns, M. N., Schueller, S. M., Clarke, G., & Klinkman, M. (2013) Behavioral intervention technologies: Evidence and future directions. *General Hospital Psychiatry*. Retrieved from <https://www.sciencedirect.com/journal/general-hospital-psychiatry>

- Naslund, J. A., Aschbrenner, K. A., Araya, R., Marsch, L. A., Unitizer, J., Patel, V., & Bartels, S. J. (2017) Digital technology for treating and preventing mental disorders in low-income and middle-income countries: A narrative review of the literature. *The Lancet Psychiatry*, 4(6), 486-500. Retrieved from [https://doi.org/10.1016/S2215-0366\(17\)30096-2](https://doi.org/10.1016/S2215-0366(17)30096-2)
- Naslund, J. A., Aschbrenner, M. C., Araya, R., Marsch, L. A., Unutzer, J., Patel, V., & Bartels, S. J. (2017). *Digital technology for mental health in low-income and middle-income countries. The Lancet Psychiatry*, 4(6), 486–500. [https://doi.org/10.1016/S2215-0366\(17\)30041-5](https://doi.org/10.1016/S2215-0366(17)30041-5)
- Norris, P. (2001) Digital divide: Civic engagement, information poverty, and the internet worldwide. Cambridge University Press.
- Norris, P. (2001). *Digital divide: Civic engagement, information poverty, and the internet worldwide* Cambridge University Press.
- Nunnally, J. C., & Bernstein, I. H. (1994) Psychometric theory (3rd ed.). McGraw-Hill.
- OECD. (2021) Digital health and mental well-being. OECD Publishing. Retrieved from <https://www.oecd.org/health/digital-health-and-mental-well-being.htm>
- Or, C. K. L., & Karsh, B. T. (2009). *A systematic review of patient acceptance of consumer health information technology. Journal of the American Medical Informatics Association*, 16(4), 550–560. <https://doi.org/10.1197/jamia.M2410>
- Penchansky, R., & Thomas, J. W. (1981). *The concept of access: Definition and relationship to consumer satisfaction. Medical Care*, 19(2), 127–140. <https://doi.org/10.1097/00005650-198102000-00001>
- Perera, S., Riddle, W. D. R., & Kumar, K. (2020). *Mental health issues in Sri Lanka: Challenges and opportunities. Journal of South Asian Studies*, 45(3), 412–428. <https://scholar.google.com/scholar?q=Perera+2020+mental+health+Sri+Lanka>

- Peterlin, B. L., Scher, A. I., Lipton, R. B., & Blumenfeld, A. M. (2023). Path analysis and structural equation modeling in headache research. *Cephalalgia*, 43(2). <https://doi.org/10.1177/03331024221142245>
- Pierce, B. S. (2020). *Telepsychiatry in the United States*. *Journal of Telemedicine and Telecare*, 26(7–8), 419–429. <https://scholar.google.com/scholar?q=Pierce+2020+telepsychiatry>
- Preacher, K. J., & Hayes, A. F. (2008) Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods*, 40(3), 879–891. Retrieved from <https://doi.org/10.3758/BRM.40.3.879>
- Riva, G., Baños, R. M., Botella, C., Wiederhold, B. K., & Gaggioli, A. (2016). *Positive technology: Harnessing the power of emotions*. *Current Opinion in Psychology*, 9, 9–14. <https://scholar.google.com/scholar?q=Riva+2016+positive+technology+emotions>
- Saunders, M., Lewis, P., & Thornhill, A. (2019) *Research methods for business students* (8th ed.). Pearson Education.
- Senanayake, B., Wickramage, K., Visvalingam, P., & Edirippulige, S. (2020) Mental health literacy and help-seeking behaviours among Sri Lankan Australians: A cross-sectional study. *BMC Public Health*, 20, 1085. Retrieved from <https://doi.org/10.1186/s12889-020-09182-7>
- Senanayake, H., & Senanayake, S. (2017). *Digital mental health in South Asia*. *Health Affairs*, 36(11), 1932–1940. <https://scholar.google.com/scholar?q=Senanayake+2017+digital+mental+health+South+Asia>

- Senanayake, M. G. B., & Senanayake, G. S. (2017) mHealth: Vehicle for health system strengthening in Sri Lanka. *Journal of Mobile Technology in Medicine*, 6(1), 41-47. Retrieved from <https://doi.org/10.7309/jmtm.6.1.5>
- Shore, J. H., Yellowlees, P. M., Caudill, R., Johnston, B., Turvey, C., Mishkind, M., Krupinski, E., Myers, K., Shore, P., Kaftarian, E., & Hilty, D. M. (2018). *Best practices in videoconferencing-based telemental health. Telemedicine and e-Health*, 24(11), 827–832. <https://doi.org/10.1089/tmj.2018.0237>
- Siriwardhana, C., Adikari, A., Pannala, G., Abas, M., Sumathipala, A., Stewart, R., & Siribaddana, S. (2013) An intervention to improve mental health care for conflict-affected forced migrants in low-resource primary care settings: A WHO MhGAP-based pilot study in Sri Lanka (COM-GAP study). *Trials*, 14(1), 423. Retrieved from <https://doi.org/10.1186/1745-621514423>
- Sundarapperuma, T. D., Gamage, M. W. K., & Rathnayake, N. (2023) Psychological disturbances encountered in Sri Lanka during COVID-19. *BMC Psychiatry*, 23, 452. Retrieved from <https://bmcpsy psychiatry.biomedcentral.com/articles/10.1186/s12888-023-04918-2>
- Telecommunications Regulatory Commission of Sri Lanka. (2023) ICT indicators 2023. TRCSL. Retrieved from <https://www.trc.gov.lk>
- Torous, J. B., Firth, J., Mueller, N., Keshavan, M., & Sardesai, S. (2019). *Emerging issues and controversies in mental health technology. JAMA Psychiatry*, 76(4), 353–354. <https://doi.org/10.1001/jamapsychiatry.2019.0093>
- Torous, J., Trevino, B., Larsen, M. E., & Firth, J. (2017). *Digital mental health and psychiatry: Raising the bar for the 21st century. World Psychiatry*, 16(3), 258–260. <https://doi.org/10.1002/wps.20446>
- Tran, T., Le, T.-H., Nguyen, N.-Q., & Tran, H.-T. (2021). Factors affecting adoption of artificial intelligence in Vietnam's banking industry. *Journal of Asian*

Finance, Economics and Business, 8(3), 1015–1025. <https://doi.org/10.13106/jafeb.2021.vol 8.no3.1015>

Venkatesh, V., & Bala, H. (2008). *Technology acceptance model 3 and a research agenda on interventions*. *Decision Sciences*, 39(2), 273–315. <https://doi.org/10.1111/j.1540-5915.2008.00192.x>

Venkatesh, V., & Davis, F. D. (2000). *A theoretical extension of the technology acceptance model: Four longitudinal field studies*. *Management Science*, 46(2), 186–204. <https://doi.org/10.1287/mnsc.46.2.186.11926>

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). *User acceptance of information technology: Toward a unified view*. *MIS Quarterly*, 27(3), 425–478. <https://doi.org/10.2307/30036540>

Wijenayake, J., Kodikara, N., & De Silva, L. (2023) Systematic literature review on AI chatbot solution for medical practitioner adoption and engagement with the healthcare system in Sri Lanka. *International Journal of Advanced Computer Science and Applications*, 14(7), 567-578. Retrieved from <https://doi.org/10.14569/IJACSA.2023.0140762>

World Health Organization. (2019). *Monitoring and evaluating digital health technologies* WHO Publications. <https://www.who.int/publications/i/item/9789241513678>

World Health Organization. (2021) *Global strategy on digital health 2020–2025*. Retrieved from <https://www.who.int/publications/i/item/9789240020924>

World Health Organization. (2022). *World mental health report: Transforming mental health for all* WHO. <https://www.who.int/publications/i/item/9789240049338>

Yamane, T. (1967) *Statistics: An introductory analysis* (2nd ed.). Harper & Row.

Appendix

Appendix 1.1



Digital Technology as a Catalyst for Improving Access to Mental Health Services in Sri Lanka

(Special Reference to Colombo Sri Lanka)

2026

Survey conducted by,

HS/2020/0633,

Department of Social Statistics, University of Kelaniya.

This questionnaire is used to collect your perspectives on digital mental health platforms, including mobile apps and telepsychiatry services, among young adults in the Colombo district. The information collected will be kept confidential and will be used for study purposes only. Your participation is highly valued, and I kindly request you to answer all questions.

Only for the data entering purpose	
Survey Dates	
Location	Colombo

Please select the suitable answer/s.

Part 1: Demographic Information

1.1 Gender			1.2 Age Group		
1	Male		1	20–24 years	
2	Female		2	25–29 years	
3	Prefer not to say		3	30–34 years	

1.3 Highest Level of Education Completed			1.4 Current Employment Status		
1	O/L or below		1	Employed (full-time)	
2	A/L		2	Employed (part-time)	
3	Diploma/Certificate		3	Self-employed	
4	Undergraduate degree		4	Student	
5	Graduate/Postgraduate degree		5	Unemployed	
1.5 Approximately, what is your monthly income?					
1	Below LKR 50,000				
2	LKR 50,000 - 100,000				
3	LKR 100,001 - 150,000				
4	LKR 150,001 - 200,000				
5	Above LKR 200,000				
6	Prefer not to answer				

Part 2: Factors

1 = Strongly Disagree

2 = Disagree

3 = Neutral

4 = Agree

5 = Strongly Agree

2.1 Digital Technology Adoption		1	2	3	4	5
1	I regularly use a smartphone or tablet to access health related information.					
2	I am aware of mobile applications specifically designed to support mental health.					
3	I have used or would be willing to use a dedicated mental health mobile application.					
4	I am familiar with telepsychiatry (remote psychiatric consultation via video or phone).					
5	I have used or would consider using an online counselling platform for mental health support.					
2.2 Barriers		1	2	3	4	5
1	I lack sufficient digital literacy skills to use mental health applications effectively.					
2	Limited or unreliable internet access in Colombo prevents me from using digital mental health services.					
3	I am concerned about the privacy and security of my personal data on mental health platforms.					
4	Social stigma around mental health in Sri Lankan culture discourages me from using digital mental health tools.					
5	The quality of available digital mental health services in Colombo is inadequate.					
6	My family opposes or resists my use of digital mental health platforms.					
7	Lack of Sinhala/Tamil language support in digital mental health tools is a barrier for me.					
8	I lack sufficient digital literacy skills to use mental health applications effectively.					

2.3 Enabling Factors		1	2	3	4	5
1	High mobile phone penetration in Sri Lanka provides a strong foundation for digital mental health services.					
2	Hospitals and clinics in Colombo actively provide digital mental health services.					
3	Public awareness campaigns about digital mental health services effectively reach young adults in Colombo.					
4	Support from family and social networks encourages me to use digital mental health platforms.					
5	The availability of Sinhala/Tamil mental health applications promotes my use of digital services.					
6	Digital mental health adoption among Colombo youth is growing rapidly.					
2.4 Access to Mental Health Services		1	2	3	4	5
1	I find it easy to access mental health services in Colombo when I need them.					
2	Digital technology has improved my ability to obtain timely mental health information.					
3	I am aware of the digital mental health services currently available in Sri Lanka.					
4	Digital mental health platforms offer more affordable support than traditional clinical services.					
5	Using digital tools, I am able to access mental health support in a timely manner.					

Part 3: Additional Preference and Open-Ended Questions

3.1: Which digital mental health tools have you used or are aware of? (Select all that apply):

- ☐ Mobile app
- ☐ Telepsychiatry
- ☐ Online counselling
- ☐ SMS helpline
- ☐ Wearable device
- ☐ None

3.2: In your opinion, what is the single biggest barrier to using digital mental health services in Sri Lanka?

3.3: What single improvement would most encourage you to use digital mental health services?

Thank you very much for your time and participation in this research!